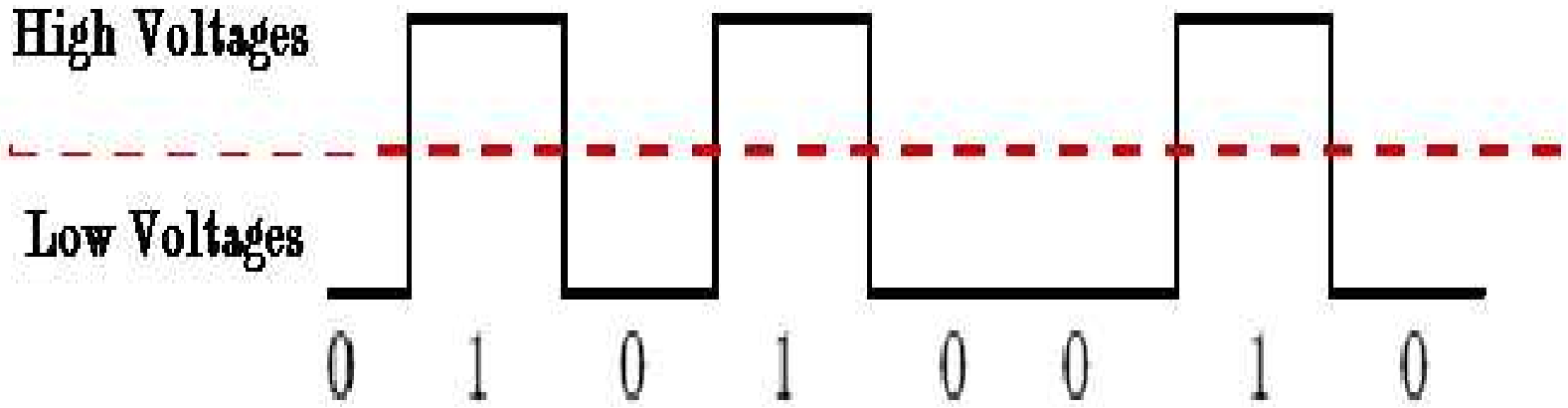
Sistem Komunikasi Digital

Materi:

1. **Pendahuluan**
2. **Modulasi Kode Pulsa (PCM)**
3. **TDM (Time Division Multiplexing)**
4. **DPCM (Differential PCM)**
5. **ADPCM (Adaptive DPCM)**
6. **Modulasi Delta**
7. **Line Code**
8. **Modulasi RF Digital**
9. **SDH (Synchronous Digital Hierarhi)**
10. **GPON (Gigabit Pasive Optical Network)**
11. **DWDM (Dense Width Division Multiplexing)**

1. Pendahuluan

- **Sistem komunikasi digital :** Suatu sistem komunikasi yang berbasis sinyal digital.
- **Sinyal digital** : sinyal data dalam bentuk pulsa yang dapat mengalami perubahan yang tiba-tiba dan mempunyai besaran 0 dan 1 (bit). Bit ”1” bisa +V (high voltage), bit “0” bisa –V(low voltage) atau 0V.



❖ Keuntungan Sitem Komunikasi Digital VS Sistem Komunikasi Analog :

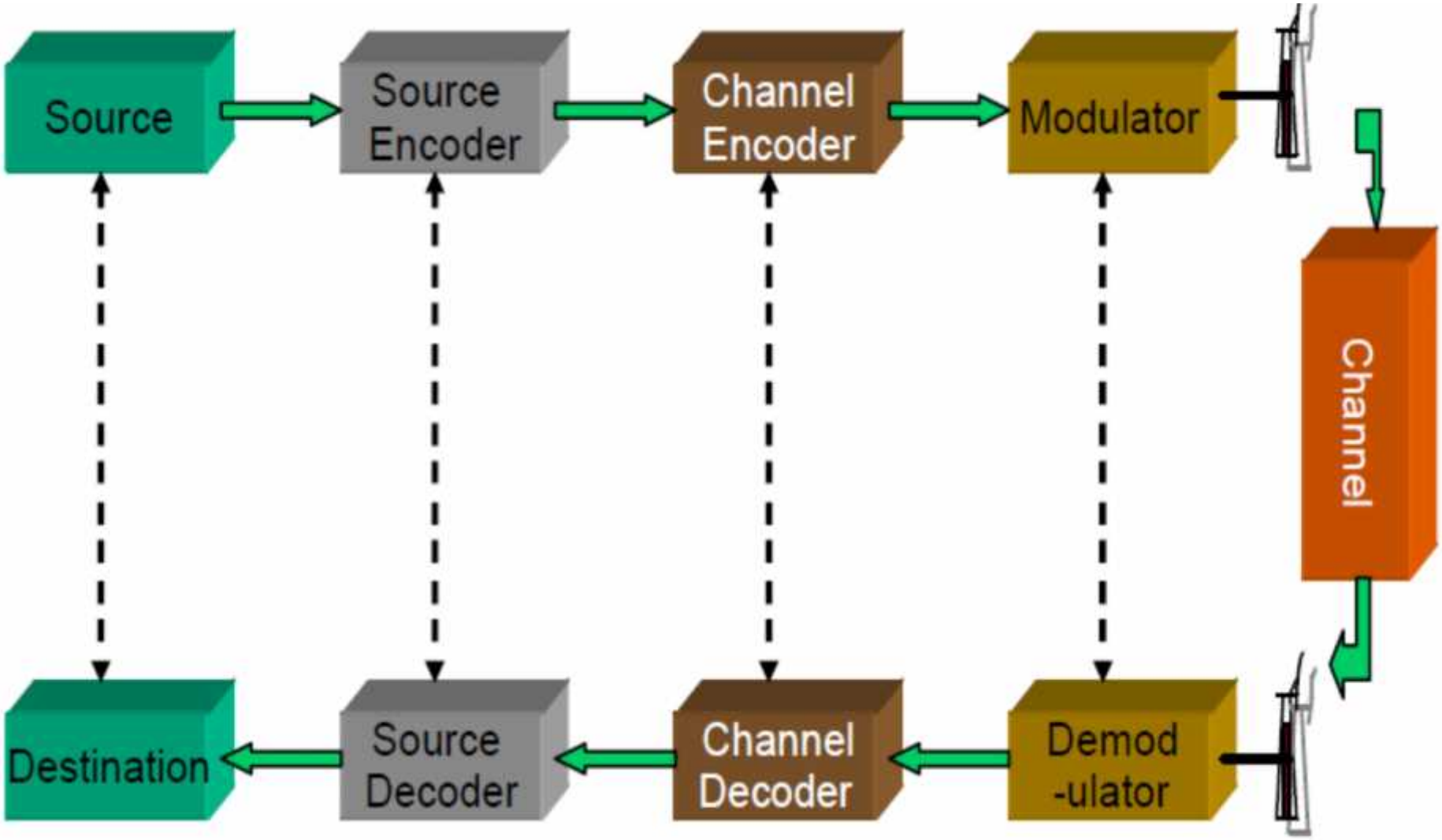
- Perencanaan rangkaian digital relatif sederhana, lebih mudah menerapkan rangkaian IC pada rangkaian digital.
- Efisiensi Biaya, komponen-komponen pada teknologi digital lebih stabil, praktis dan memiliki daya tahan lama dalam pemakaiannya, hal ini menyebabkan biaya pemeliharaan menjadi lebih sedikit. Teknologi digital menawarkan biaya lebih rendah, keandalan (reliability) yang lebih baik, pemakaian ruang yang lebih kecil, dan konsumsi daya yang rendah.
- Keutuhan data pada saat proses transmisi, sinyal digital dapat dibentuk /dibangkitkan kembali selama transmisi, kemampuan kode sinyal digital untuk meminimumkan/menekan pengaruh noise dan interferensi, dan teknologi digital lebih toleran terhadap noise.

- Makin bertambahnya penggunaan dan tersedianya teknik-teknik pengolahan sinyal digital.
- Teknologi digital memungkinkan pengenalan layanan-layanan baru, sesuai dengan teknologi komputer masa kini, teknologi digital menawarkan fleksibiltas, teknologi ISDN.
- Teknologi digital menyediakan kapasitas transmisi yang besar.
- Kemudahan Enkripsi data.

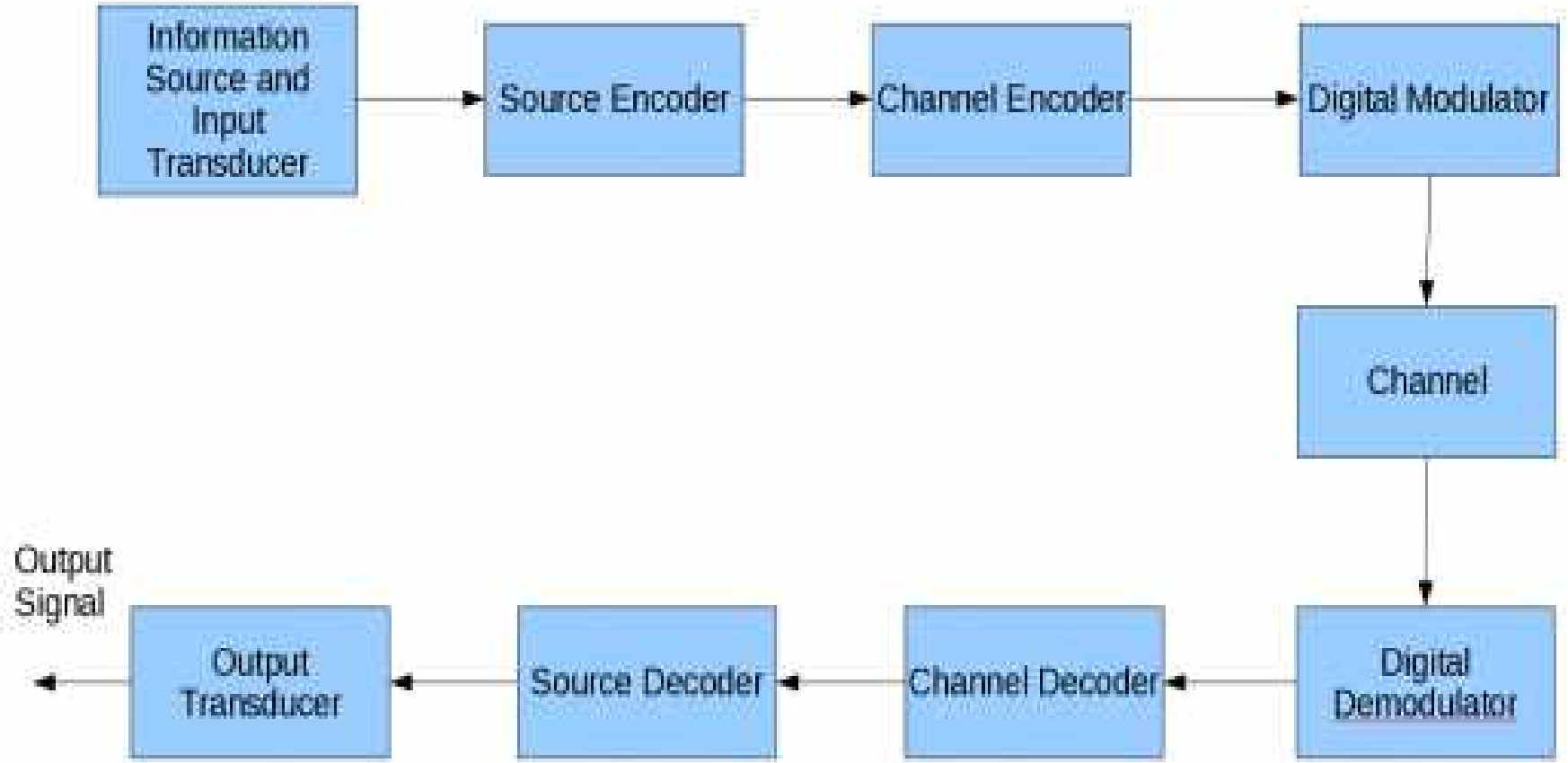
❖**Keterbatasan/Kekurangan Sitem Komunikasi Digital VS Sistem Komunikasi Analog :**

- Bandwidth relatif lebih besar/lebar.
- Terjadinya quantization error (kesalahan kuantisasi).
- Diperlukan clock yang akurat pada saat terjadi diproses coding dan decoding, perlu sinkronisasi antara pengirim dengan penerima.
- Saat ini harga komponen relatif lebih mahal.

Typical Communication System



➢ ELEMEN – ELEMEN DASAR SISTEM KOMUNIKASI DIGITAL



Basic elements of Digital Communication System

- **Information Source and Input Transducer:**

  The source of information can be analog or digital, e.g. analog: audio or video signal, digital: like teletype signal. In digital communication the signal produced by this source is converted into digital signal consists of 1′s and 0′s. For this we need source encoder.

- **Source Encoder**

  In digital communication we convert the signal from source into digital signal as mentioned above. The point to remember is we should like to use as few binary digits as possible to represent the signal. In such a way this efficient representation of the source output results in little or no redundancy. This sequence of binary digits is called ***information sequence.***

  Source Encoding or Data Compression: the process of efficiently converting the output of wither analog or digital source into a sequence of binary digits is known as source encoding.

- **Channel Encoder:**

  The information sequence is passed through the channel encoder. The purpose of the channel encoder is to introduced, in controlled manner, some redundancy in the binary information sequence that can be used at the receiver to overcome the effects of noise and interference encountered in the transmission on the signal through the channel.

  e.g. take k bits of the information sequence and map that k bits to unique n bit sequence called code word. The amount of redundancy introduced is measured by the ratio *n/k* and the reciprocal of this ratio (k/n) is known as *rate of code or code rate.*

- **Digital Modulator:**

  The binary sequence is passed to digital modulator which in turns convert the sequence into electric signals so that we can transmit them on channel (we will see channel later). The digital modulator maps the binary sequences into signal wave forms , for example if we represent 1 by sin x and 0 by cos x then we will transmit sin x for 1 and cos x for 0. (a case similar to BPSK)

- **Channel:**

  The communication channel is the physical medium that is used for transmitting signals from transmitter to receiver. In wireless system, this channel consists of atmosphere , for traditional telephony, this channel is wired , there are optical channels, under water acoustic cahnenls etc.

  we further discriminate this channels on the basis of their property and characteristics, like AWGN channel etc.

- **Digital Demodulator:**

  The digital demodulator processes the channel corrupted transmitted waveform and reduces the waveform to the sequence of numbers that represents estimates of the transmitted data symbols.

- **Channel Decoder:**

  This sequence of numbers then passed through the channel decoder which attempts to reconstruct the original information sequence from the knowledge of the code used by the channel encoder and the redundancy contained in the received data

  The average probability of a bit error at the output of the decoder is a measure of the performance of the demodulator – decoder combination. THIS IS THE MOST IMPORTANT POINT,

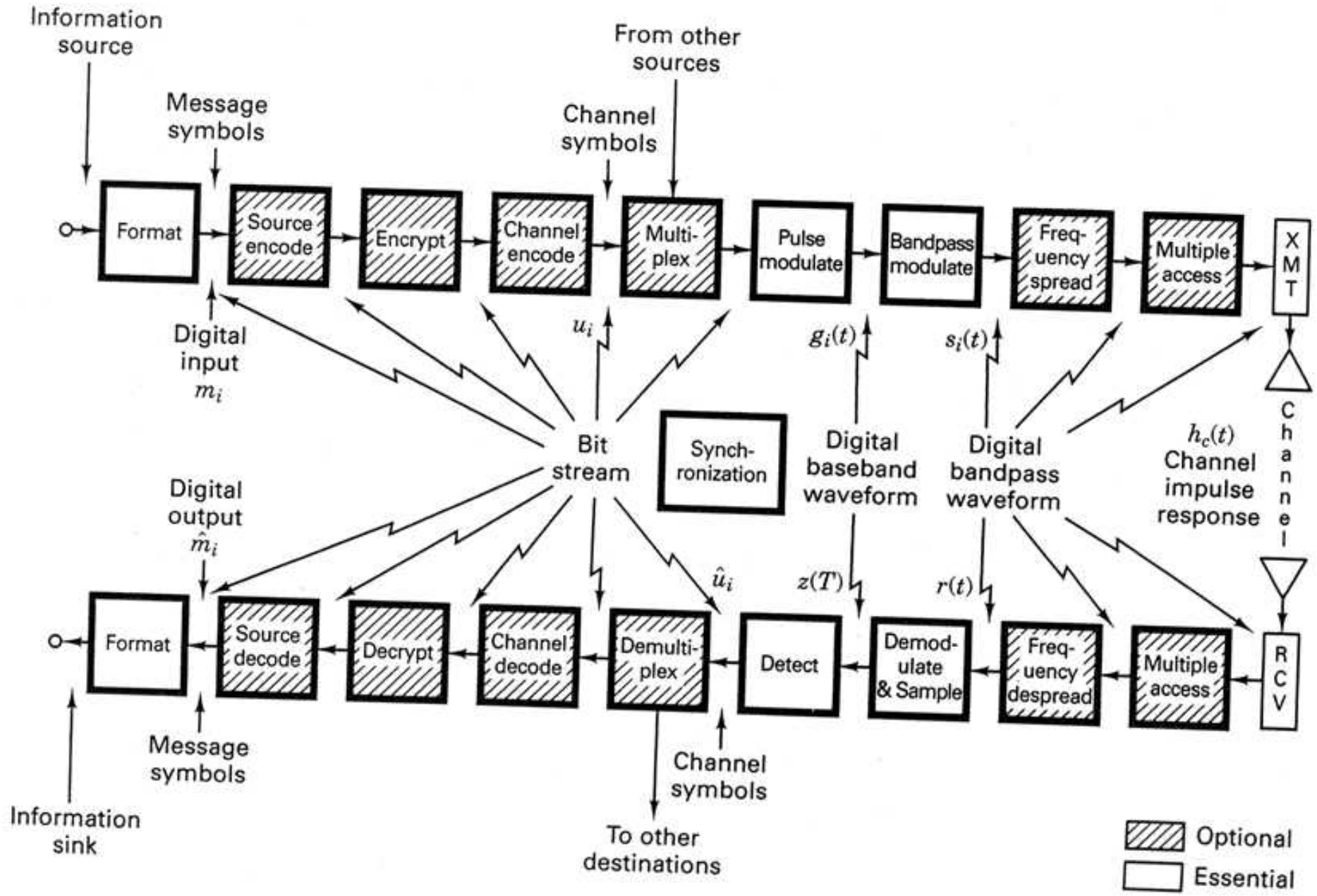
- **Source Decoder**

  At the end, if an analog signal is desired then source decoder tries to decode the sequence from the knowledge of the encoding algorithm. And which results in the approximate replica of the input at the transmitter end

- **Output Transducer:**

  Finally we get the desired signal in desired format analog or digital.

➢ ELEMEN – ELEMEN DASAR SISTEM KOMUNIKASI DIGITAL LENGKAP



2. **Modulasi Kode Pulsa** (PCM = Pulse Code Modulation)

- Salah satu teknologi memproses sinyal informasi analog menjadi sinyal digital yang ekivalen.
- Proses-proses pada sistem PCM :
  - di sisi pengirim :
    1. Sampling (pencuplikan)
    2. Quantizing (kuantisasi)
       a. Kuantisasi linier
       b. Kuantisasi non-linier
    3. Coding (pengkodean)
  - di sisi penerima :
    4. Decoding (pengkodean kembali)

- **Proses sampling :**
  - “ Mencari sample-sample dari beberapa informasi dengan menentukan titik-titik yang mewakili”.

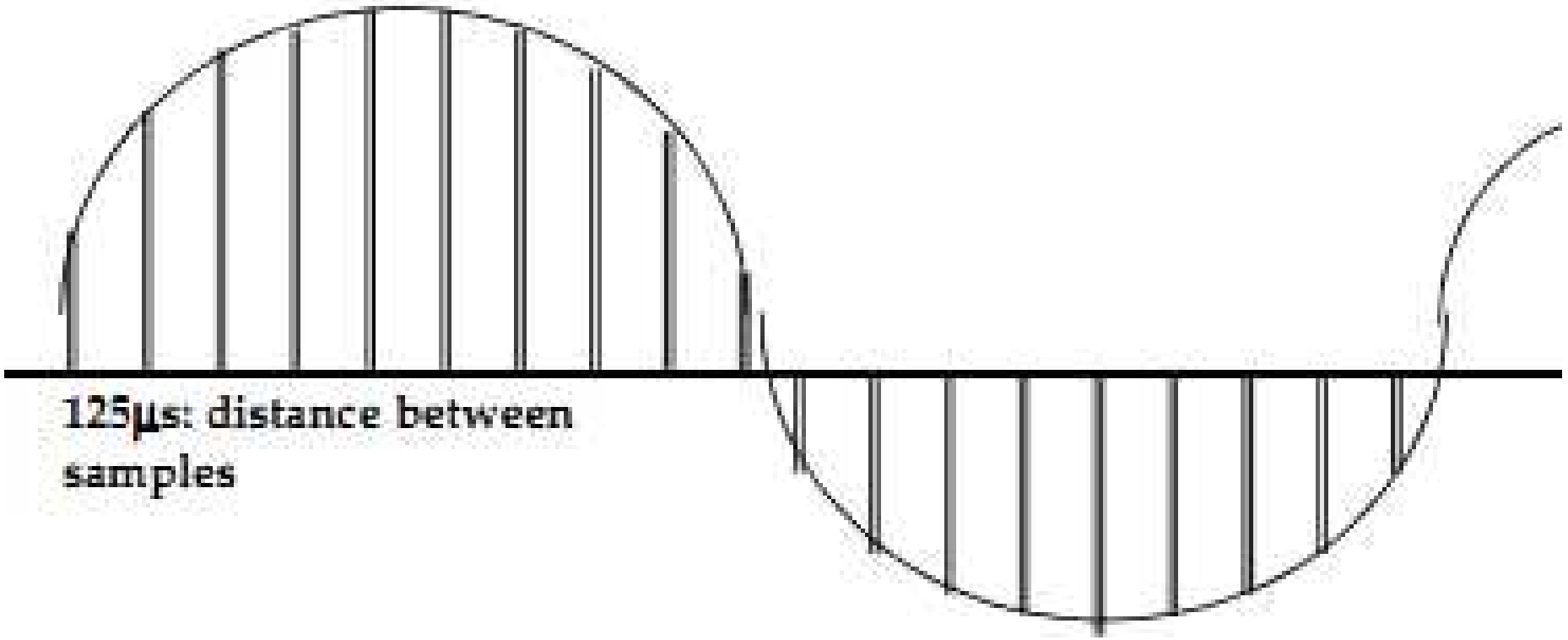
  - **Nyquist theorem :**

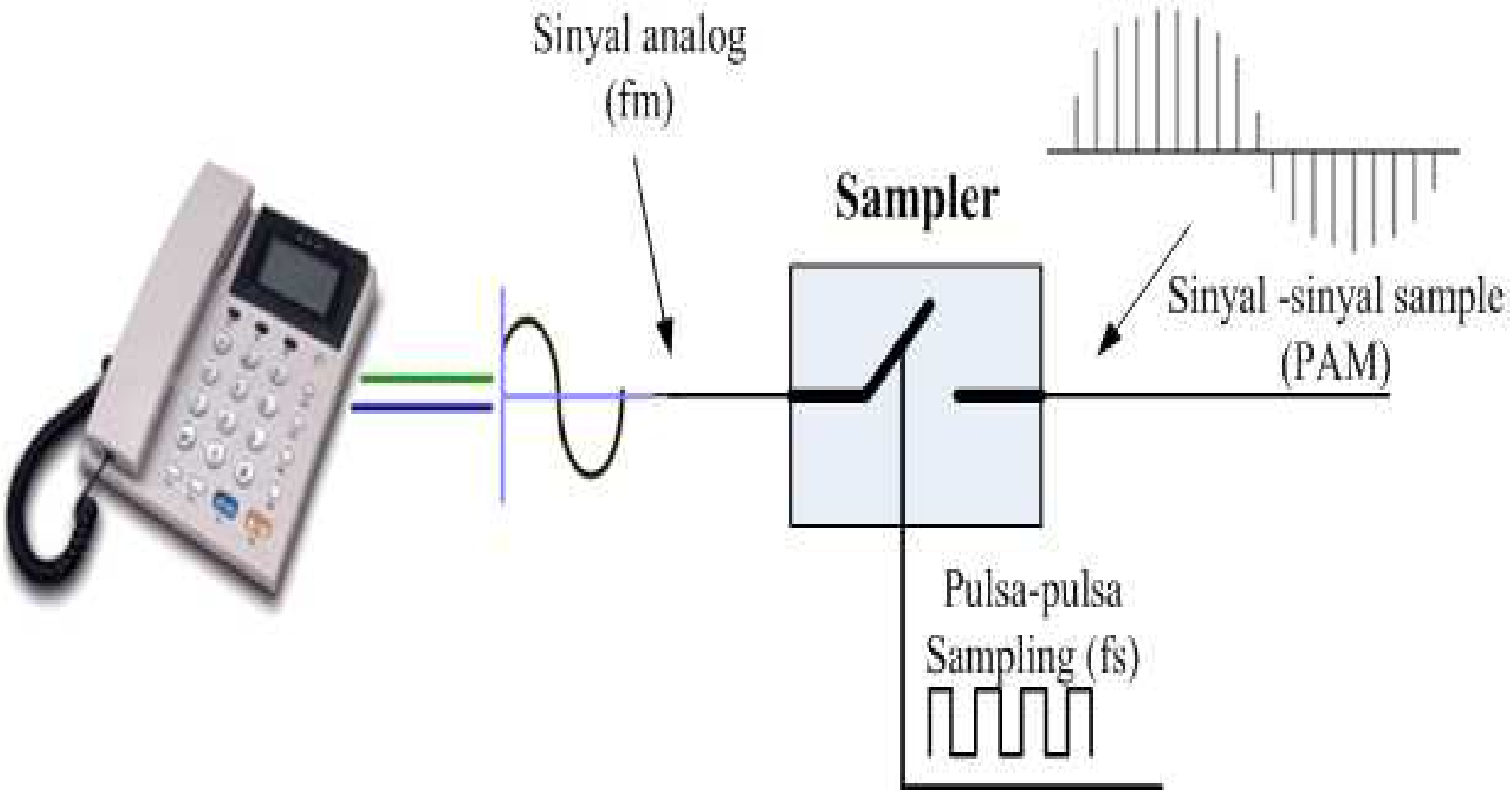
    “*Sebuah sinyal informasi analog dengan spektrum terbatas di cuplik (sampled) secara beraturan dengan frekuensi pencuplikan paling tidak 2x dari frekuensi maksimum Sinyal informasi analog tersebut, sinyal informasi analog tersebut dapat direkonstruksi kembali dengan sebuah low pass filter*”.

$$f_s \geq 2\, f_m$$

  - fs = frekuensi sampling
    fm = frekuensi maksimum sinyal informasi analog

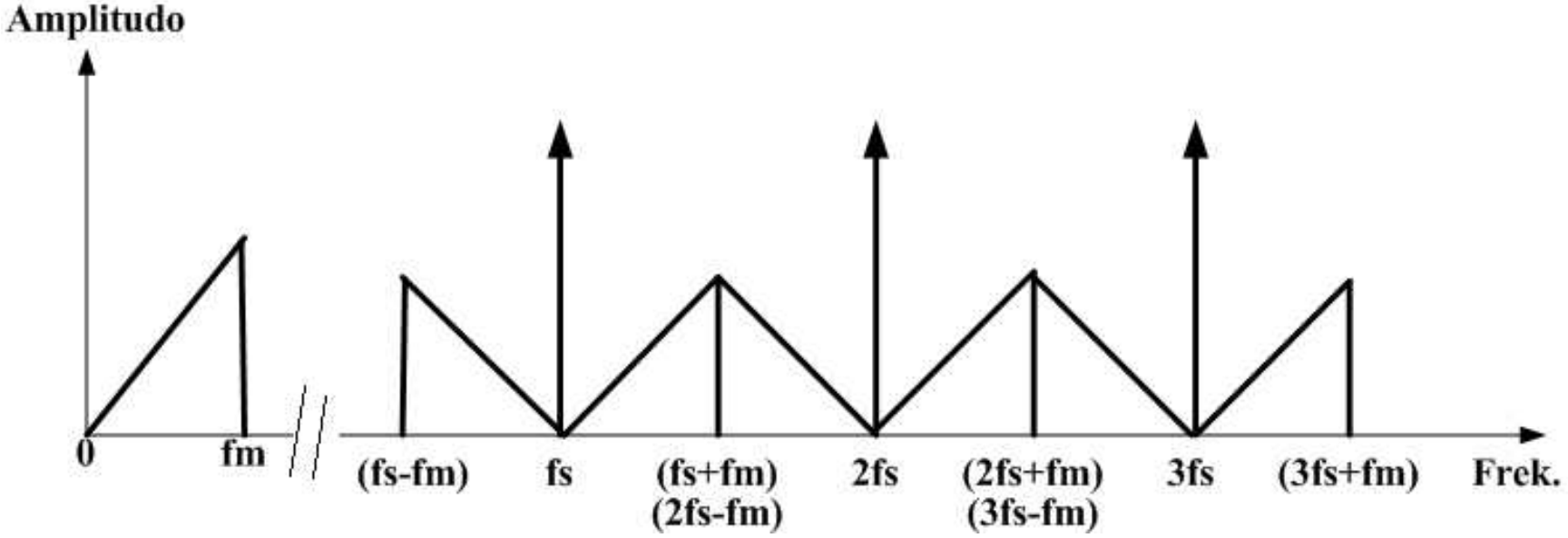
- Proses sampling menghasilkan sinyal diskret PAM yg. mencerminkan amplitudo sinyal analog.
- Jika bandwidth frekuensi sinyal informasi analog (voice) :0 s/d 3.4 Khz (dibulatkan menjadi 4 Khz, maka frekuensi pulsa sampling sebesar 8 Khz.
- waktu yang dibutuhkan dari satu sinyal sample ke sinyal sample berikutnya sebesar 125 **µS**



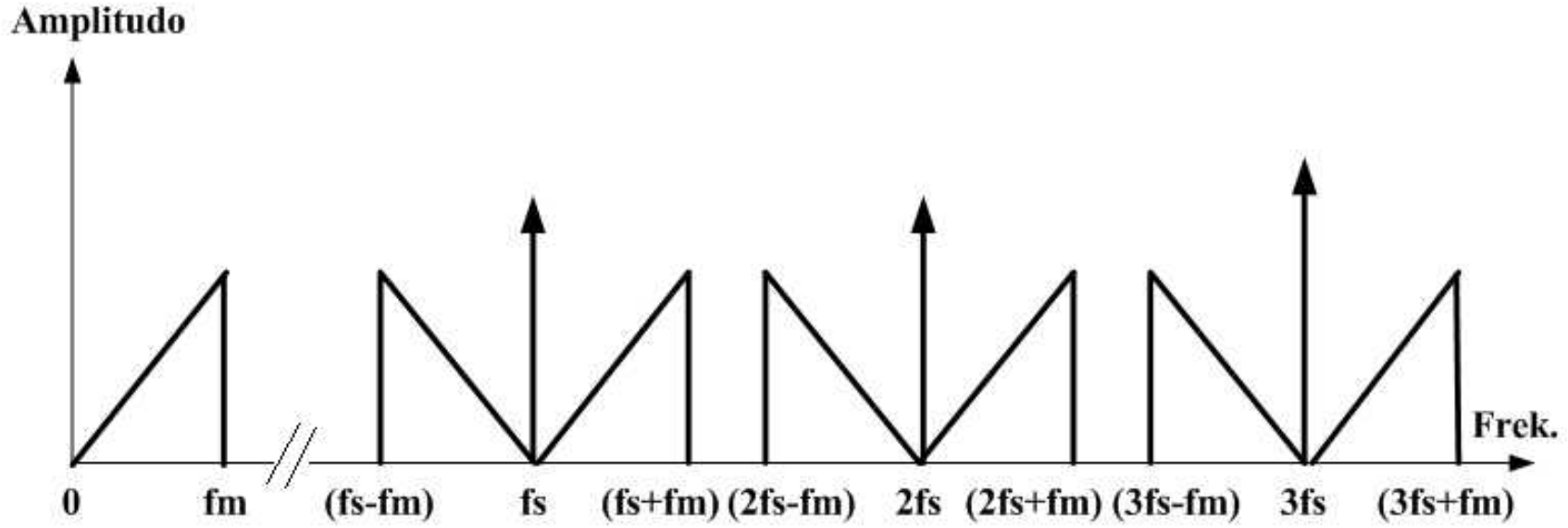


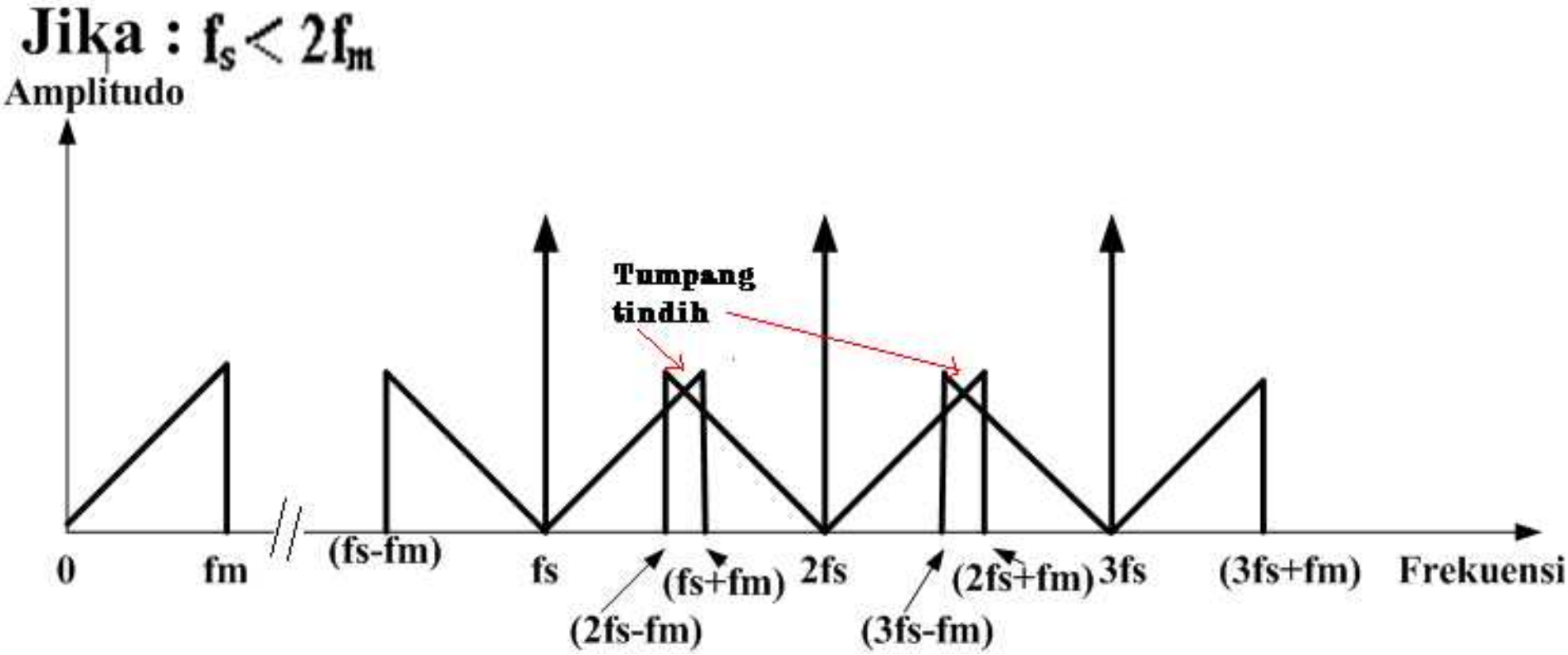
Proses sampling

Jika : $f_s = 2f_m$



Jika : $f_s > 2f_m$

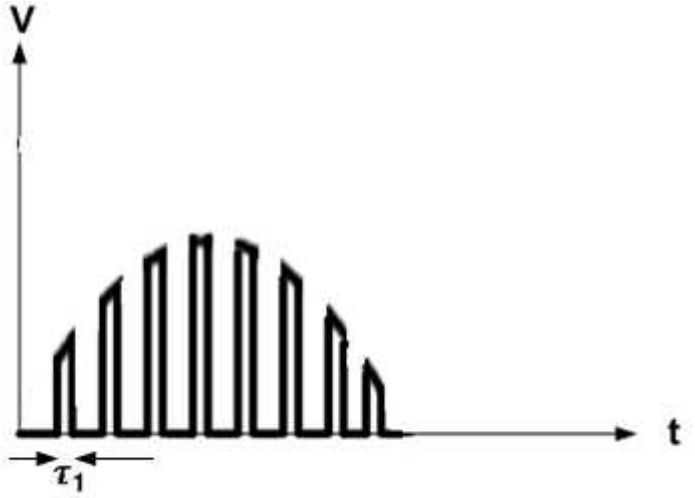




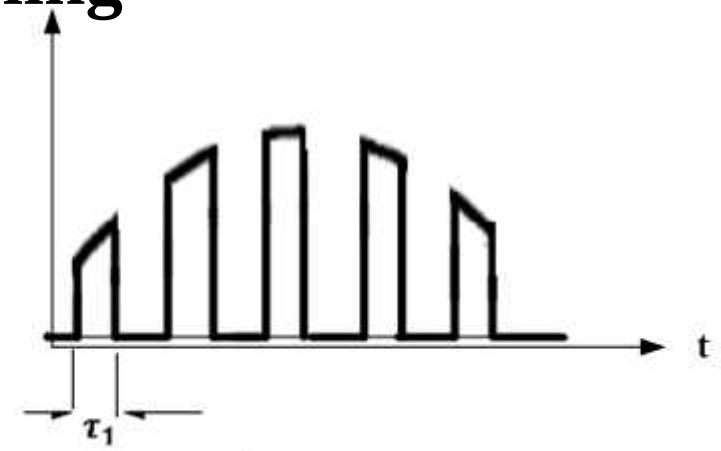
- Jika frekuensi sampling lebih kecil dari 2 x frekuensi sinyal informasi analognya, maka akan terjadi tumpang tindih pada spektrum sinyal PAM.

➢Macam-macam Metoda Sampling

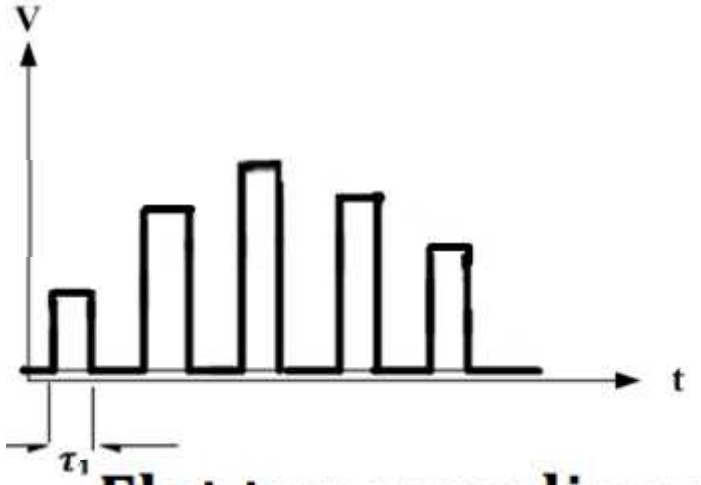
1. Instantinous Sampling
2. Natural Sampling
3. Flat Top sampling



Instantinous Sampling τ_1
sempit



Natural sampling τ_1
lebih lebar



Flat top sampling τ_1
lebih lebar

❖ Proses Kuantisasi dan Coding :

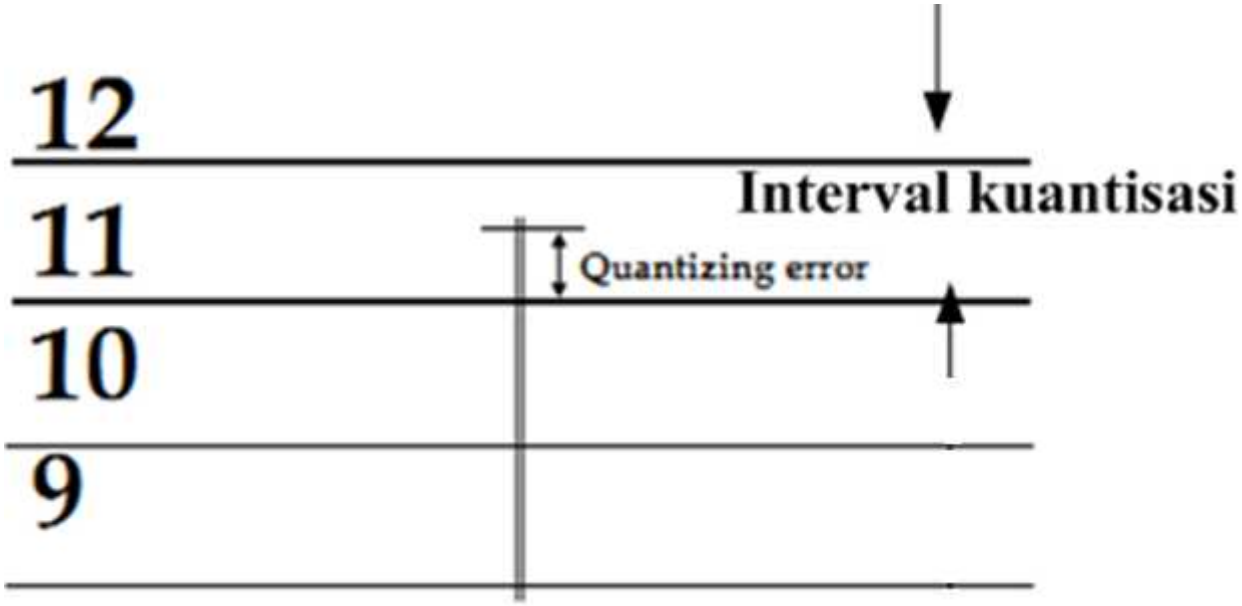
1. Kuantisasi Linier (uniform)
2. Kuantisasi non-linier (non-uniform)

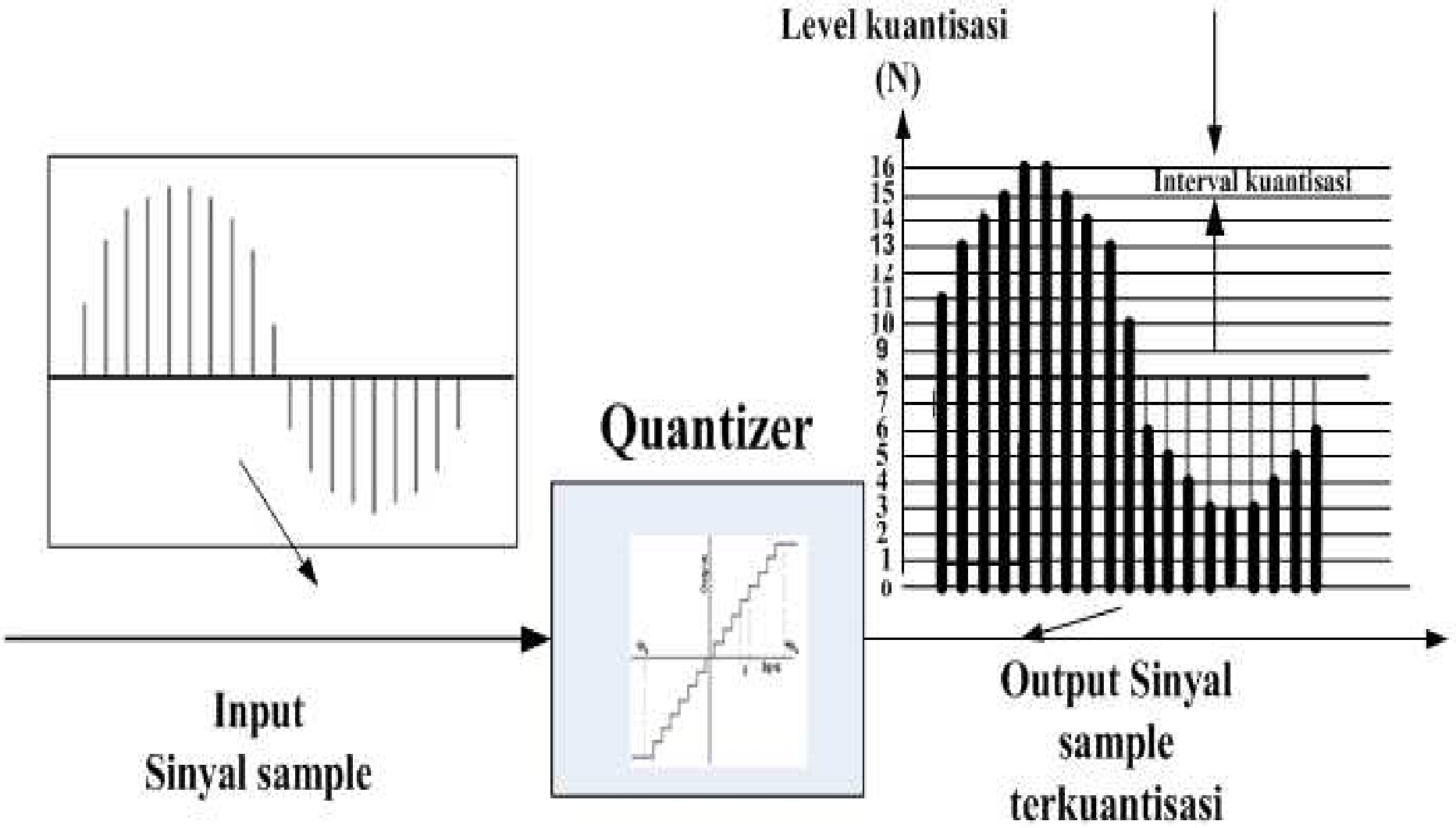
Kuantisasi Linier :

- Sinyal-sinyal PAM output dari sampler diberikan ke rangkaian quantizer untuk diberi level tertentu .
- Sinyal PAM dikuantisasi untuk menghasilkan kode-PCM
- Kuantisasi = penggantian harga riel oleh integer terdekat

  Misal, sinyal PAM ke 1 diberi level 11
  sinyal PAM ke 2 diberi level 13
  sinyal PAM ke 3 diberi level 14, dst.
- Jarak antara satu level disebut sebagai **Interval Kuantisasi**.
- Sinyal output quantizer merupakan sinyal **PAM terkuantisasi**.

- Terjadi **distorsi kuantisasi** atau **kesalahan kuantisasi** (quantization error), kesalahan kuantisasi maksimum sebesar ½ **dari interval kuantisasi.**
  - Misal sinyal PAM dengan amplitudo 10,5 diberi level 11
    sinyal PAM dengan amplitudo 11 juga diberi level 11
    sinyal PAM dengan amplitudo 11,4 diberi juga level 11
    sinyal PAM dengan amplitudo 11,6 diberi level 12
    dst.





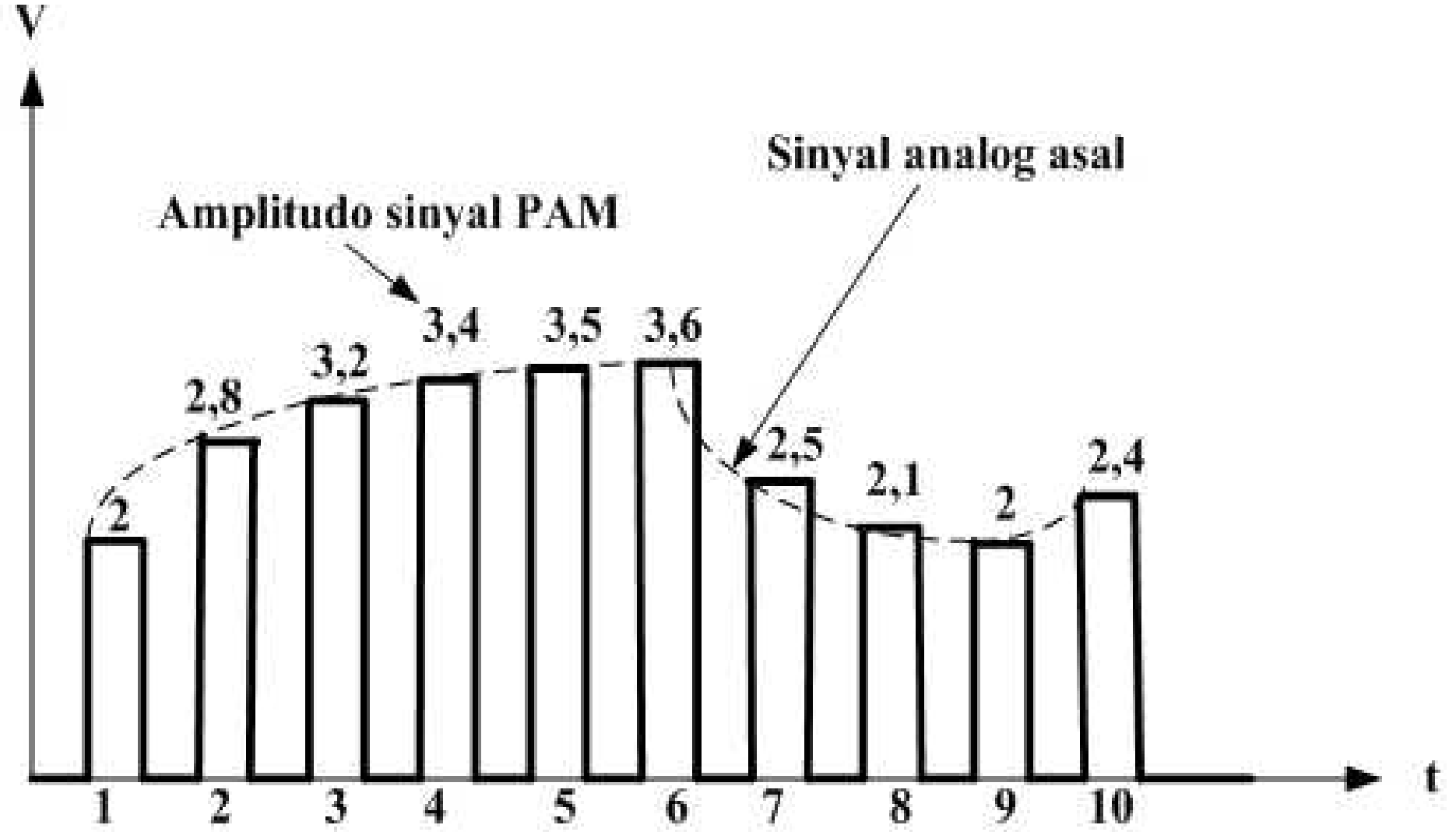
Proses Kuantisasi Linier

- Untuk Kuantisasi Linier, perbandingan antara daya sinyal dengan daya distorsi kuantisasi adalah :

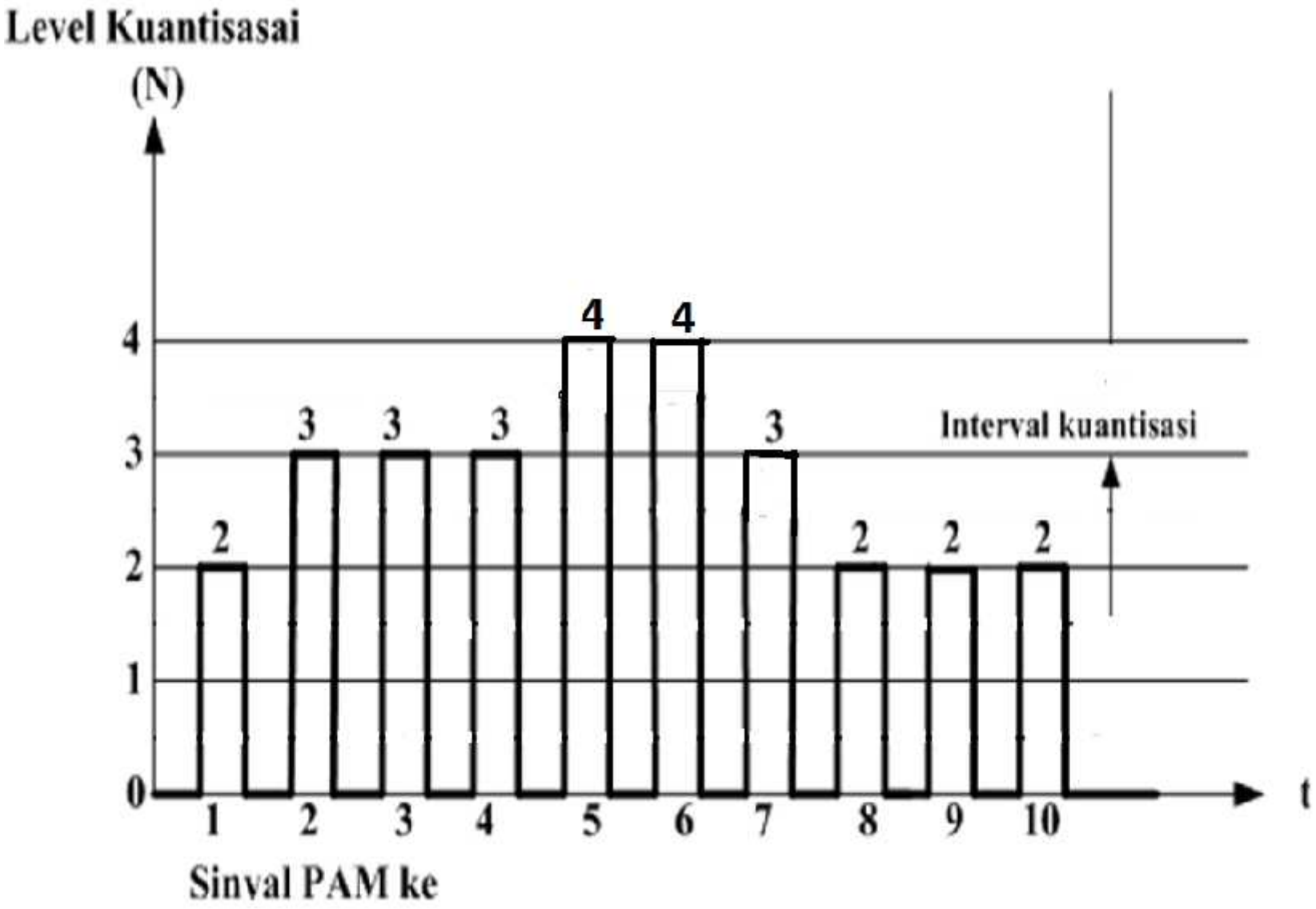
$$S/D = 6n + 1{,}8 \text{ dB} \quad n = \text{word length}$$

- Untuk memperkecil distorsi kuantisasi dapat dilakukan dengan cara :
    1. **Memperbesar level kuantisasi / memperkecil interval kuantisasi**
    2. **Kuantisasi non-linier (non-uniform)**

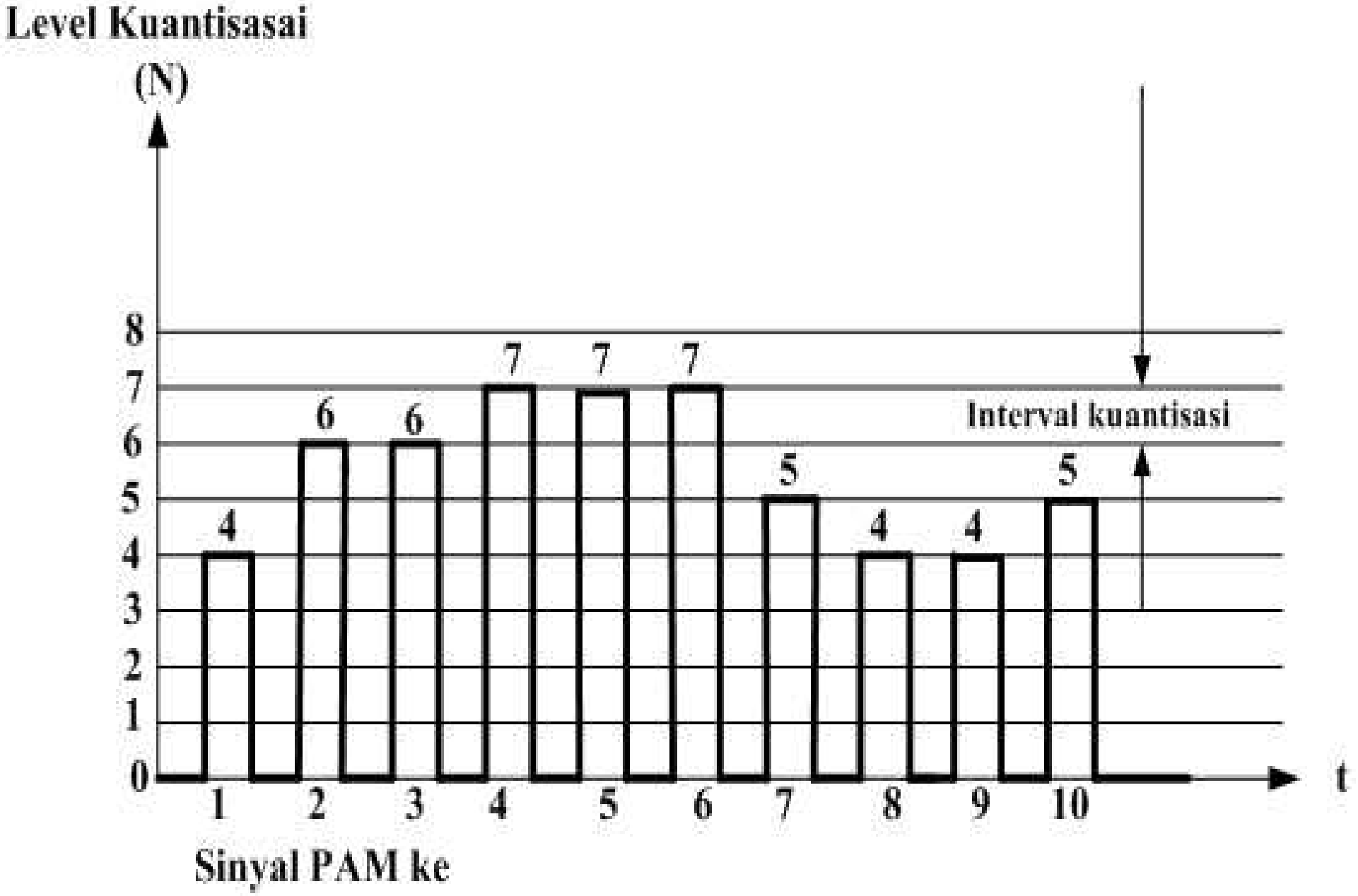
- **Memperbesar Level Kuantisasi / memperkecil Interval Kuantisasi**



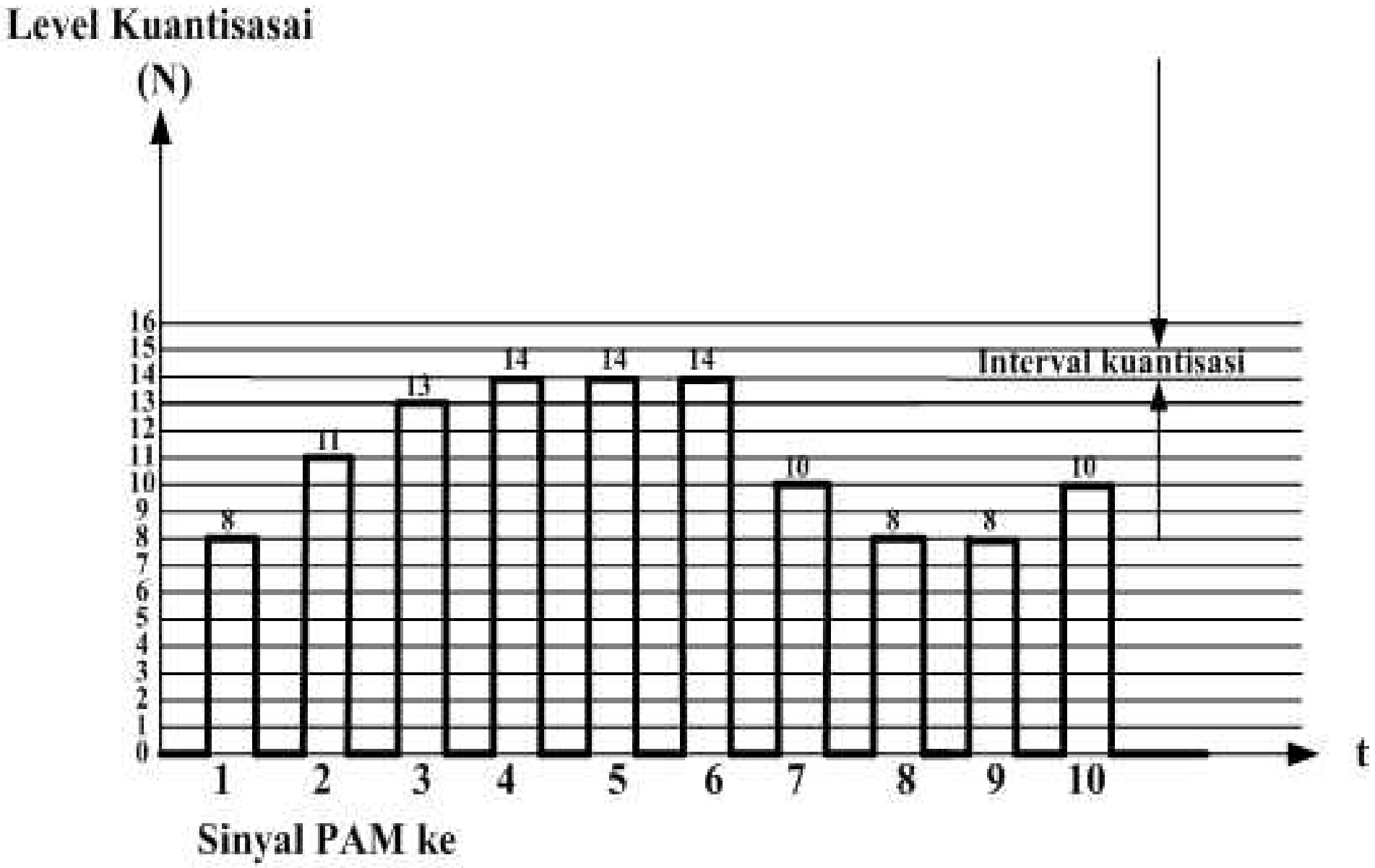
Sinyal PAM sebelum dikuantisasi



Sinyal PAM terkuantisasai dengan N = 4



Sinyal PAM terkuantisasai dengan N = 8

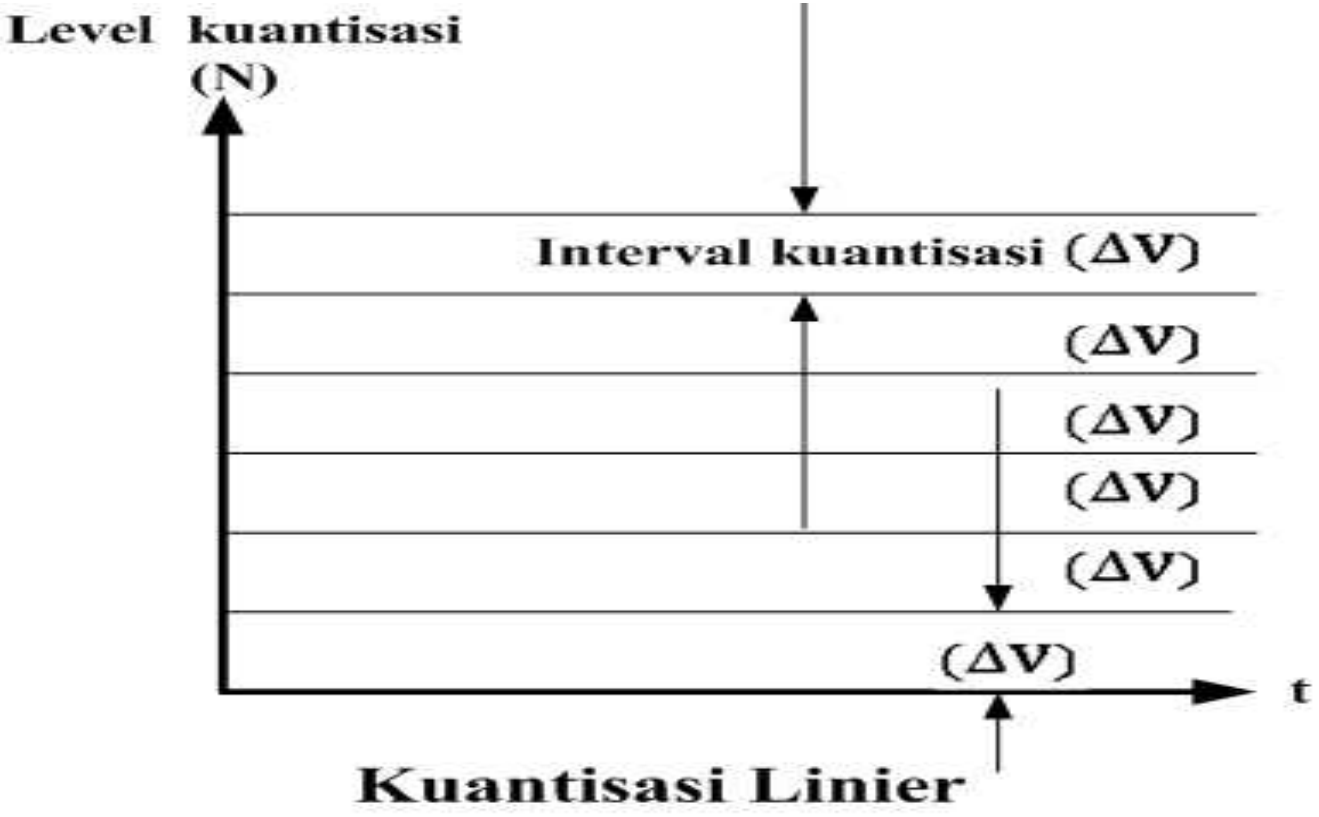


Sinyal PAM terkuantisasai dengan N = 16

▪ Kuantisasi Linier Vs Kuantisasi Non-Linier :

➢ **Kuantisasi linier :**

- Interval kuantisasi berharga konstan untuk setiap level kuantisasi, sehingga harga distorsi kuantisasi hampir sama untuk setiap level sinyal PAM, baik sinyal berlevel kecil maupun berlevel besar, sehingga S/Dq untuk sinyal PAM berlevel rendah menjadi ***"kecil",*** dan S/Dq untuk sinyal PAM berlevel tinggi menjadi ***"besar".***



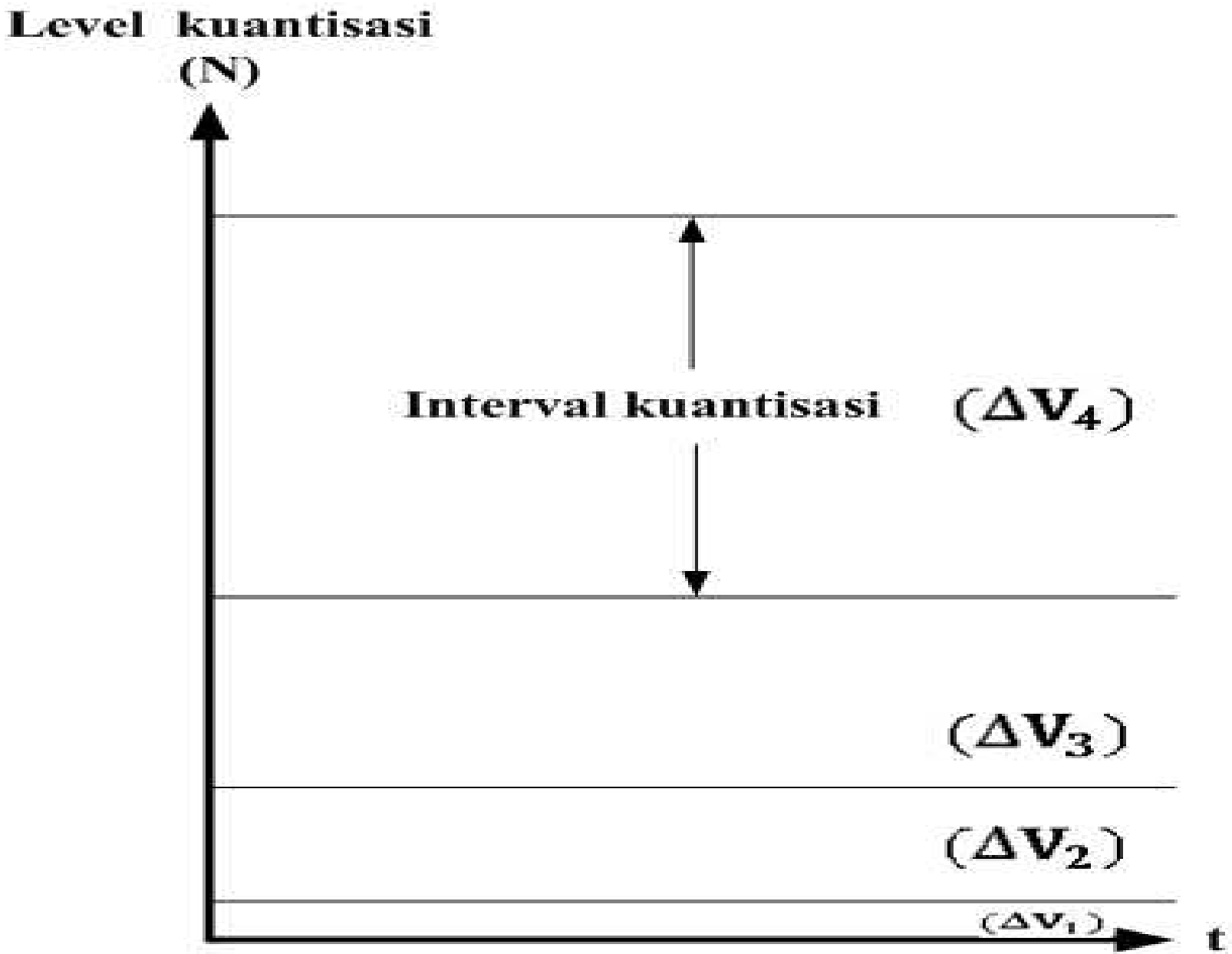
- Sistem komunikasi yang baik jika S/Dq untuk "***sinyal berlevel kecil*** " ≅ S/Dq ***"sinyal berlevel besar"***

- Dengan ***kuantisasi linier*** dapat diperoleh kualitas sistem yang ***baik*** jika ***level kuantisasi*** sebesar ***2000*** atau ***satu (1) sinyal sample/word*** (PAM terkuantisasi) dikodekan menjadi ***11 bit***.., ***boros bandwidth !!!!***

- Maka digunakanlah **"Kuantisasi non-linier"**

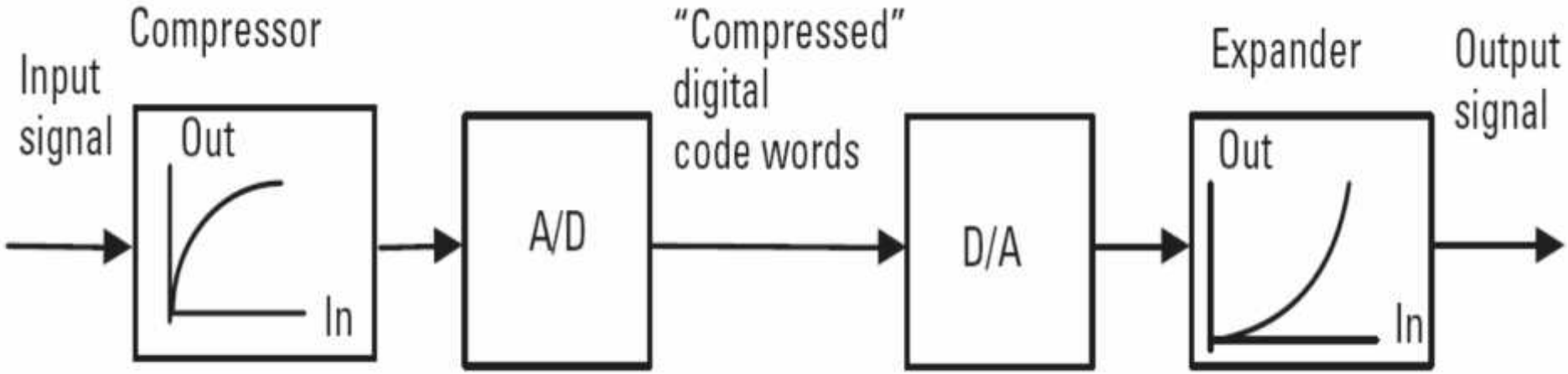
➢ **Kuantisasi Non-Linier :**

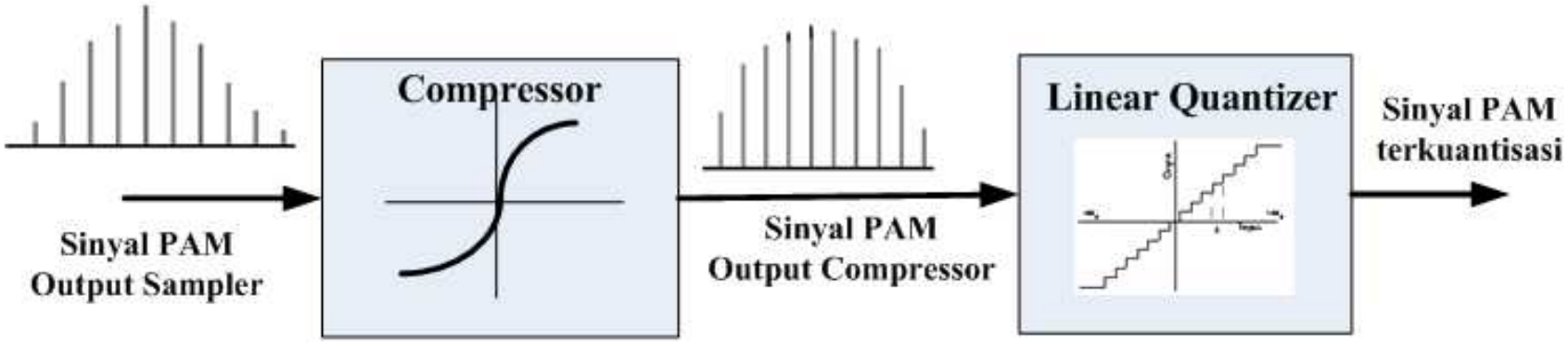
- Jarak antara interval kuantisasi tidak linier / tidak sama (logaritmis)
- Cukup diperlukan 128 level kuantisasi atau identik dengan 1 word (1 Sample sinyal PAM) dikodekan menjadi 7 bit
- Rekomendasi CCITT : Rec.G 711, kuantisasi non-linier dengan level 256 atau ekivalen dengan 1 word (1 Sample sinyal PAM) dikodekan menjadi 8 bit



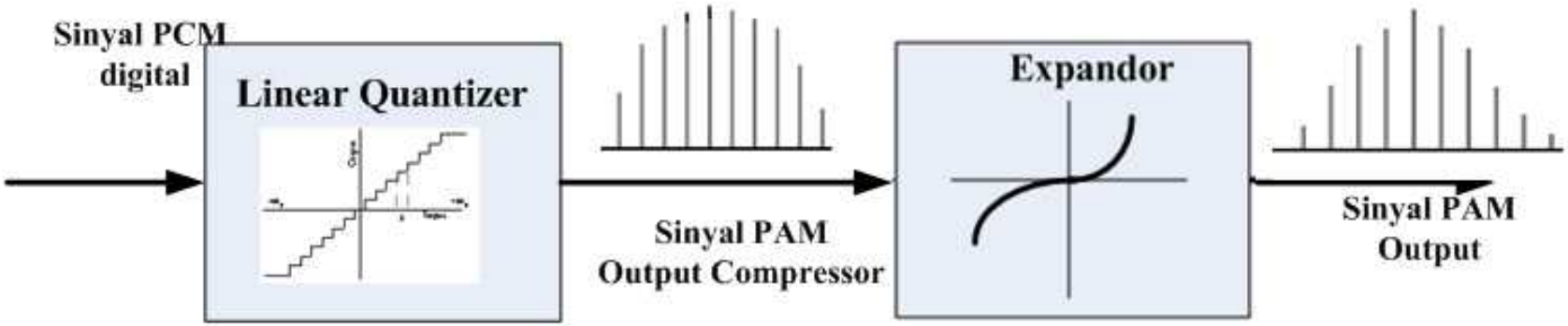
Kuantisasi Non-Linier

- Pada kuantisasi non-linier terjadi proses **Companding** (kompresi dan ekspansi).
- Pada kompresi: sinyal berlevel lebih rendah dikuatkan lebih tinggi dari pada sinyal berlevel tinggi.
- Pada proses ekspansi : sinyal berlevel lebih rendah diredam lebih besar dari pada sinyal berlevel tinggi.
- Ada 2 macam proses kompanding
  1. A law (Eropa, Indonesia)
  2. μ law (Amerika, Jepang)

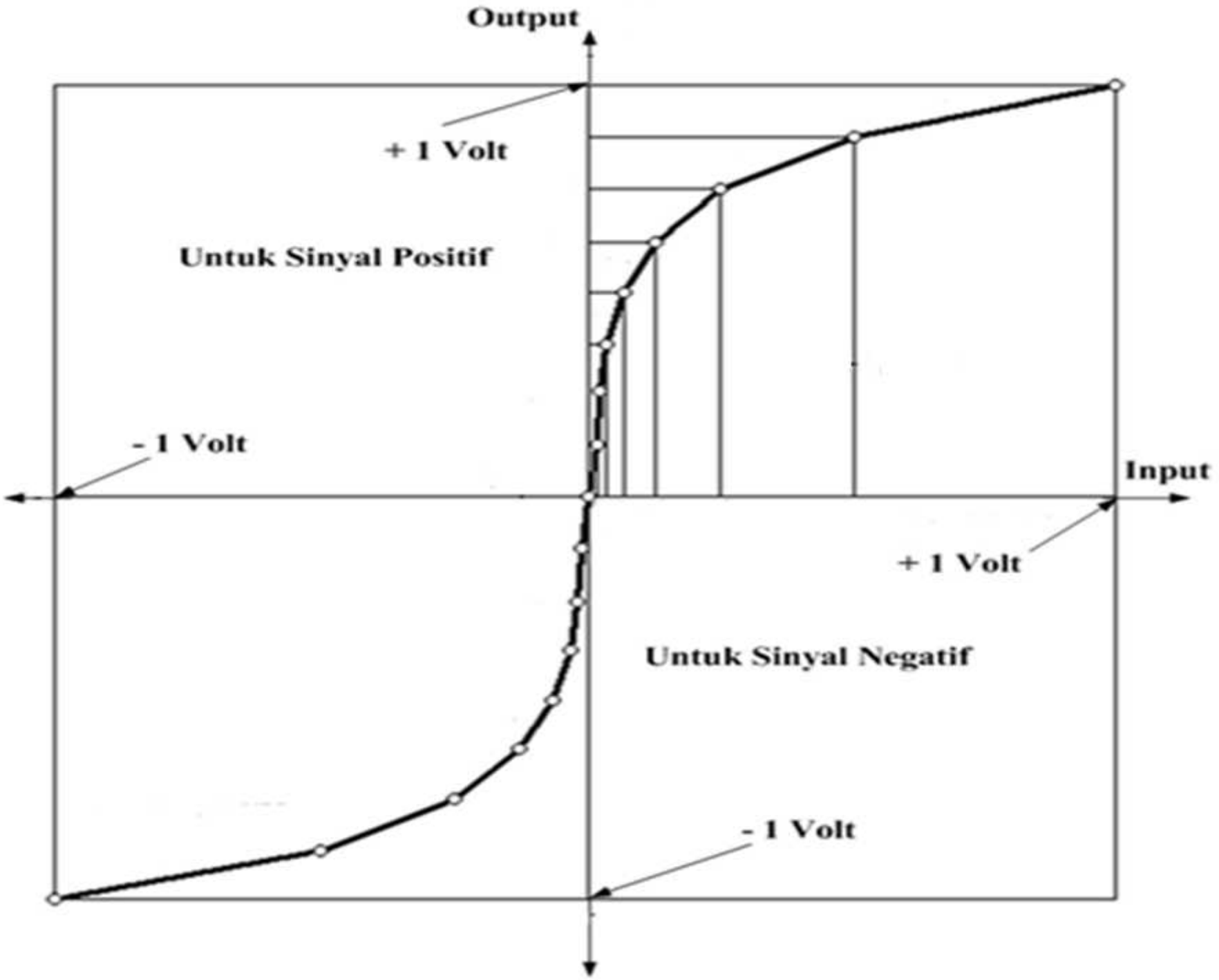


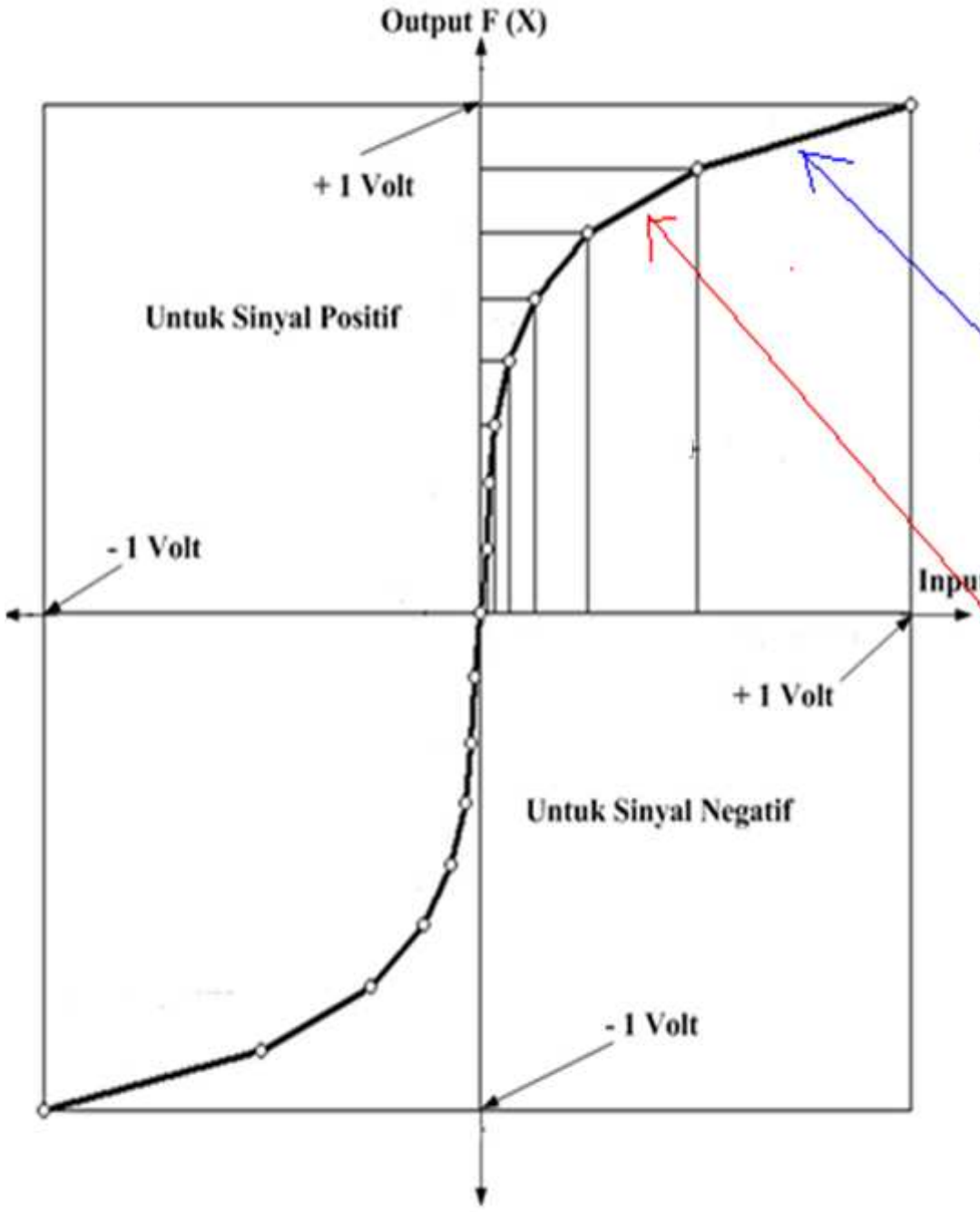


Proses Kompresi (Pengirim)



Proses Ekspansi (Penerima)


Output
+ 1 Volt
Untuk Sinyal Positif
- 1 Volt
Input
+ 1 Volt
Untuk Sinyal Negatif
- 1 Volt



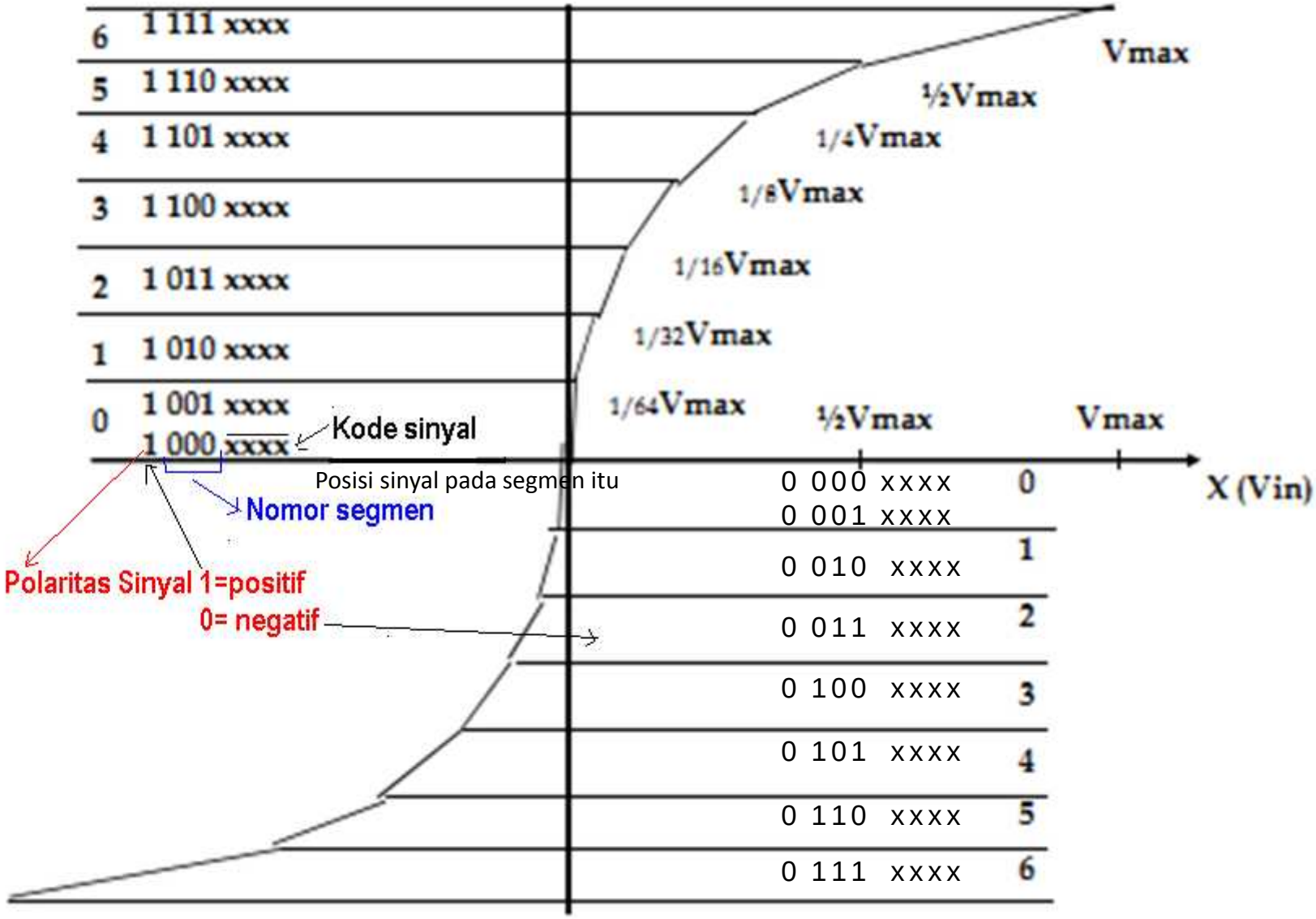
$$F(X) = \left(\frac{A|x|}{1+\ln(A)}\right), 0 \le |x| \le \frac{1}{A}$$

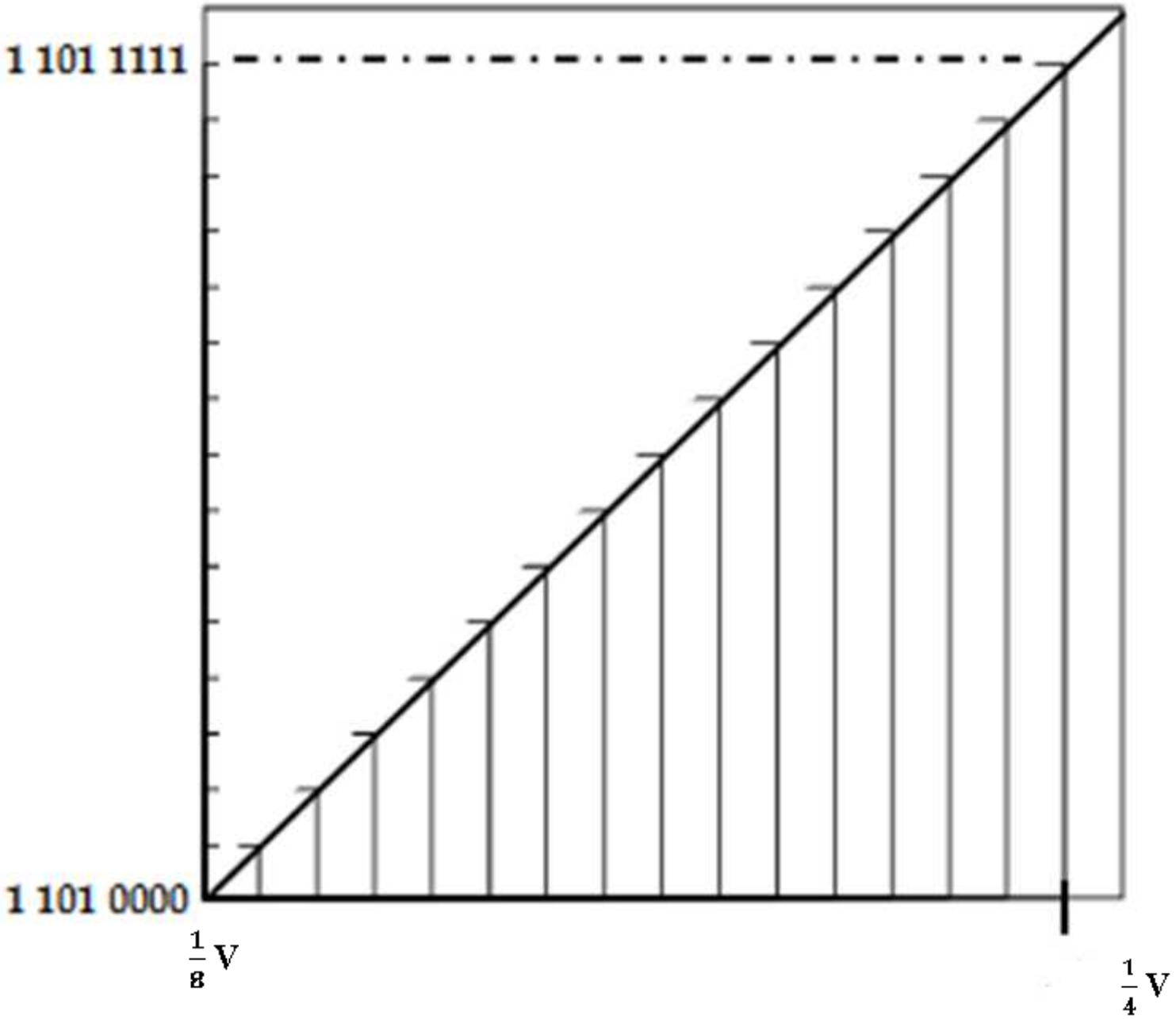
$$F(X) = \left(\frac{1+\ln|Ax|}{1+\ln(A)}\right), \frac{1}{A} \le |x| \le 1$$

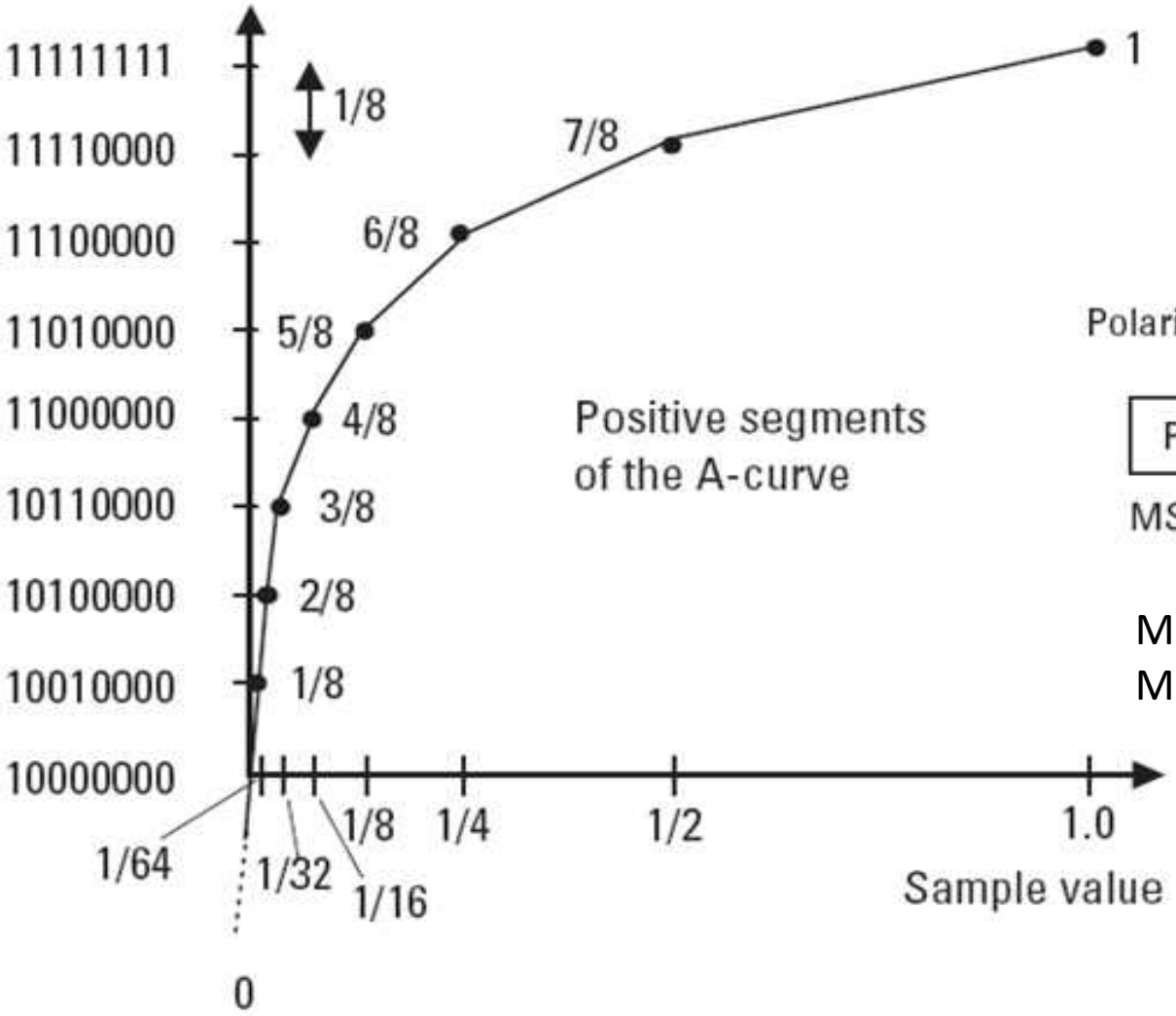
The value of A is 87,6.

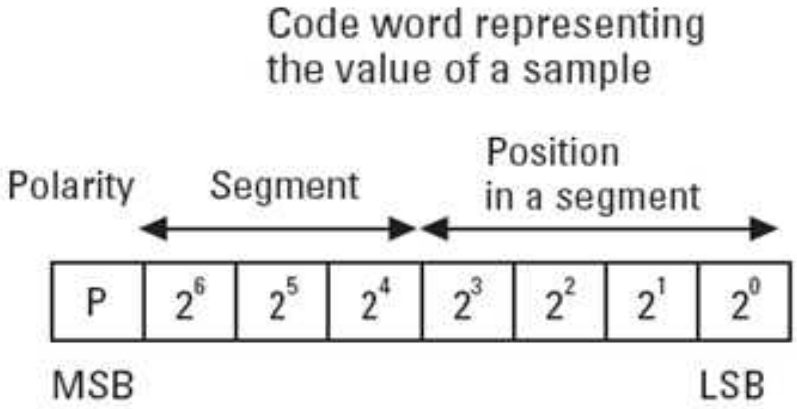
$$F(X) = \frac{\ln(1+\mu|x|)}{\ln(1+\mu)}$$

Quantizing according to the A-law

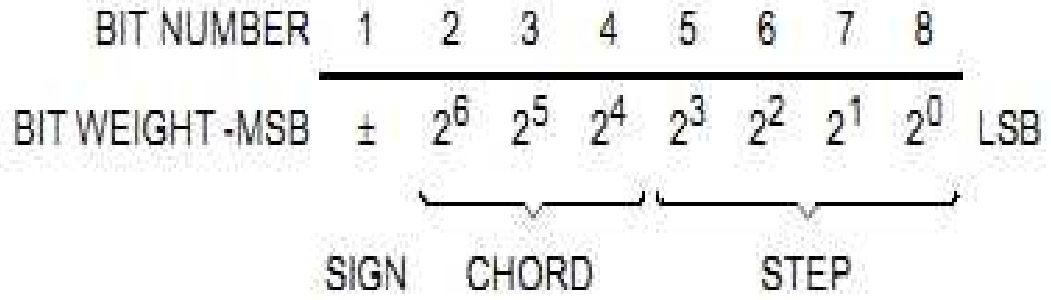


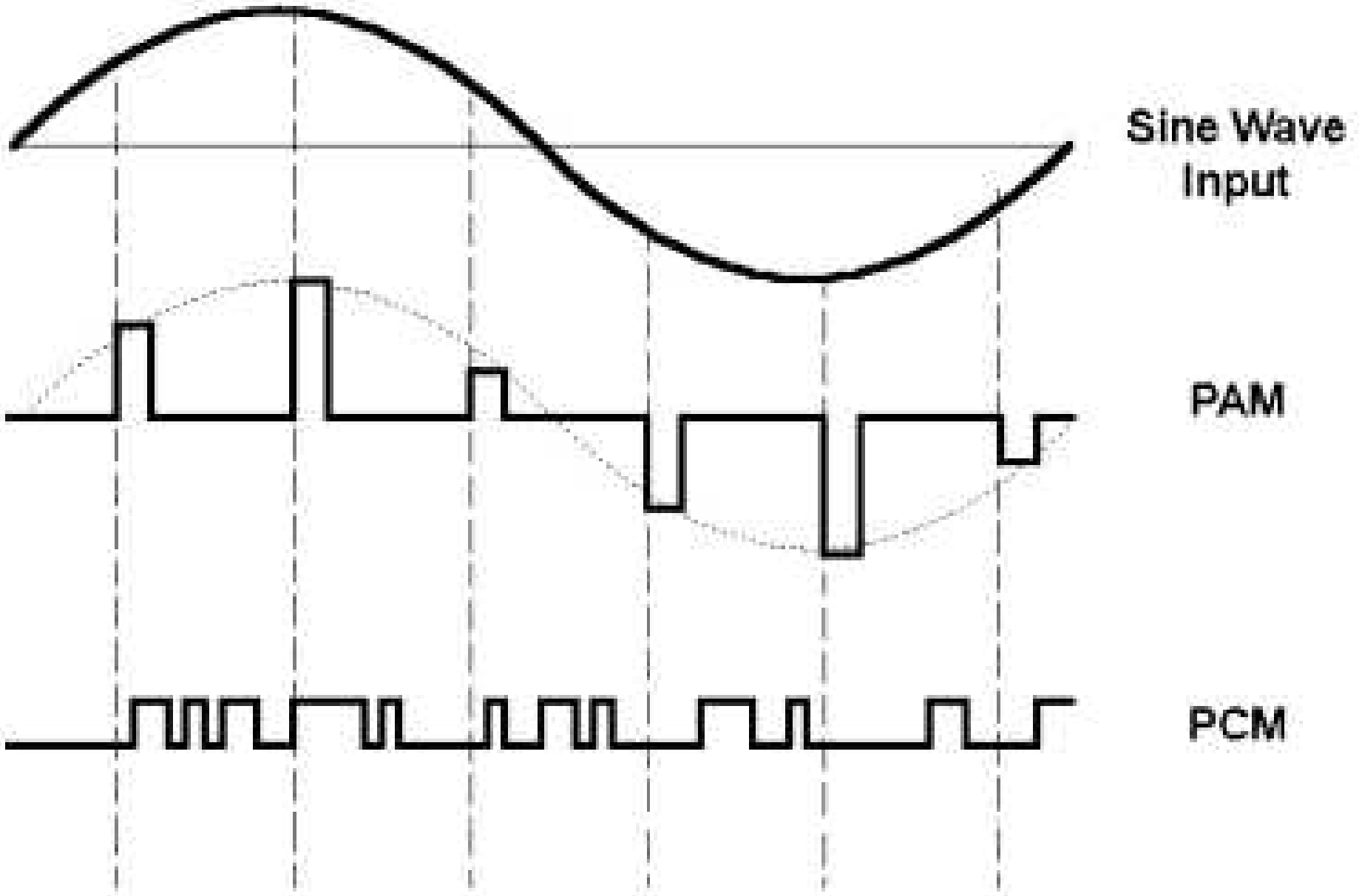

1 101 1111
1 101 0000
$\frac{1}{8}$V
$\frac{1}{4}$V

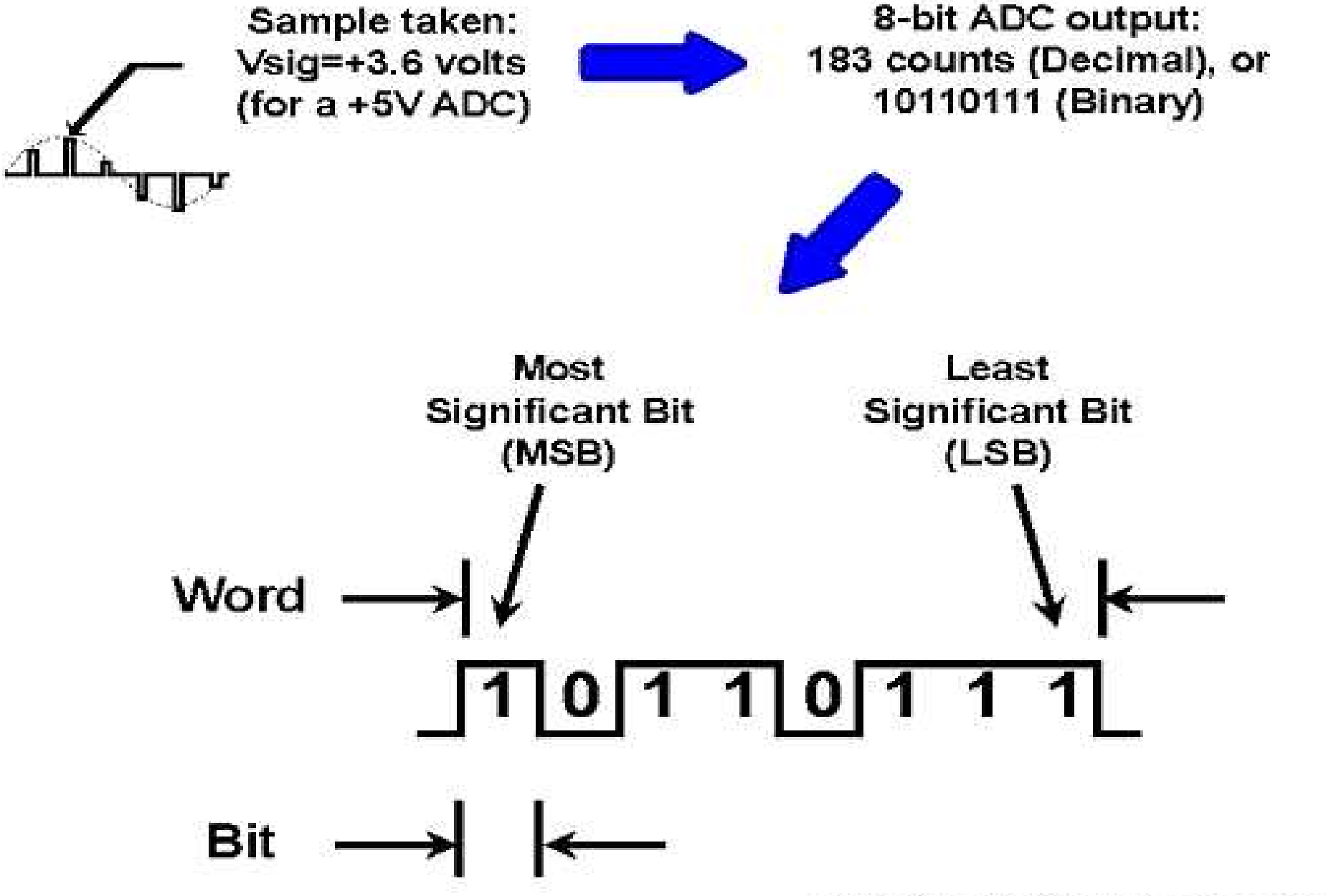




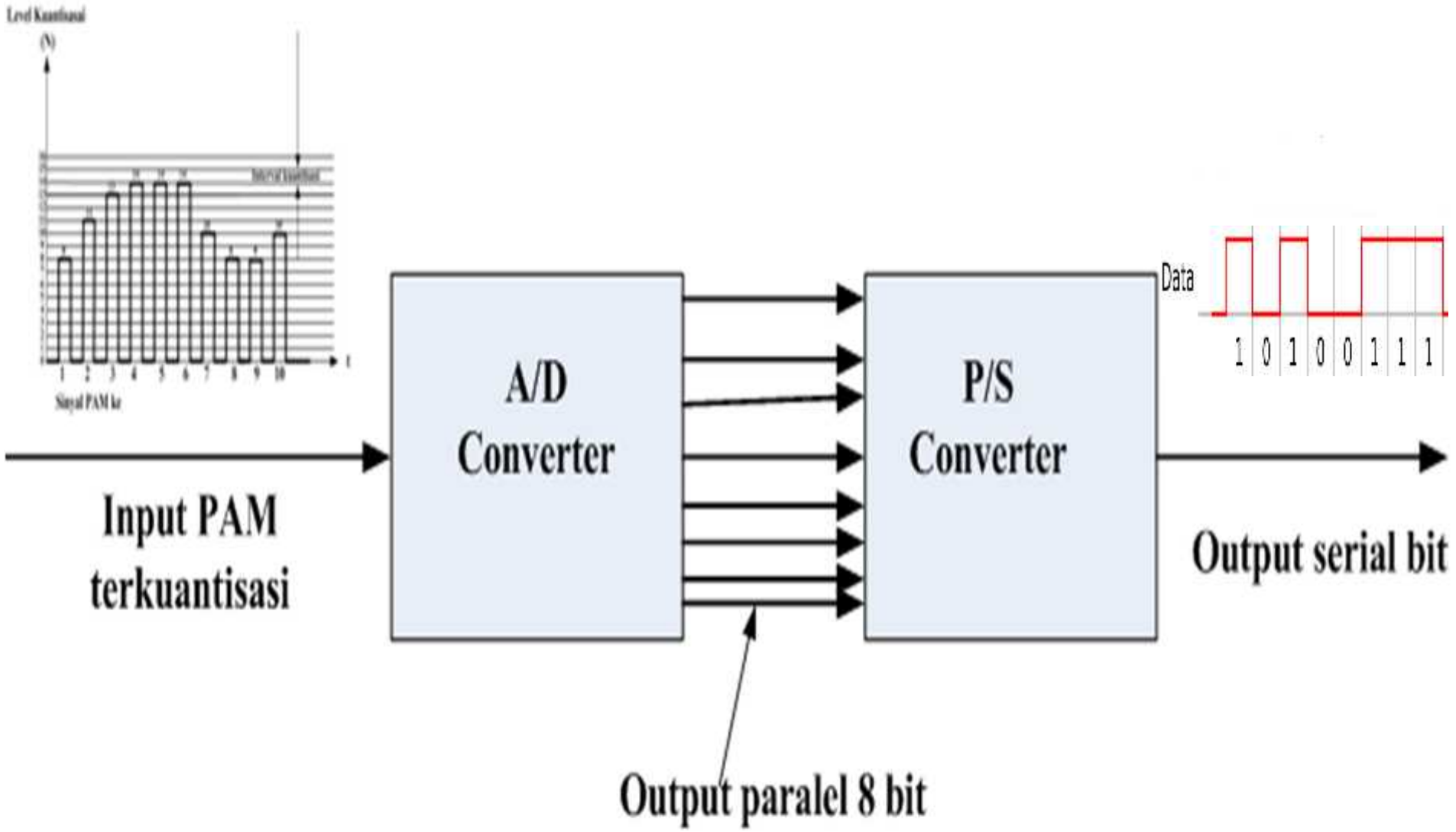
MSB = 1 untuk Sinyal Positif
MSB = 0 untuk sinyal negatif



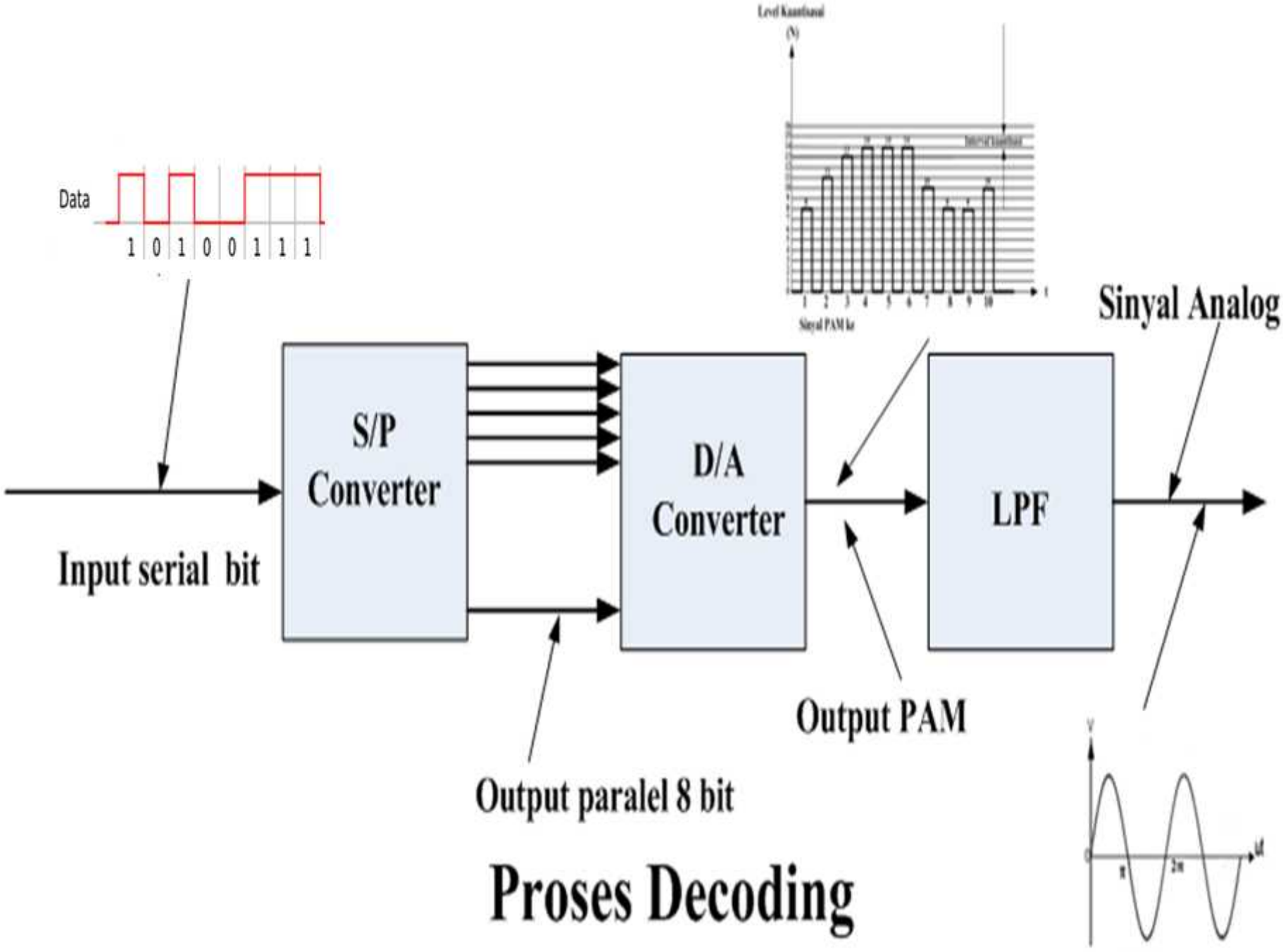

Sine Wave
Input
PAM
PCM



PCM Word Format example

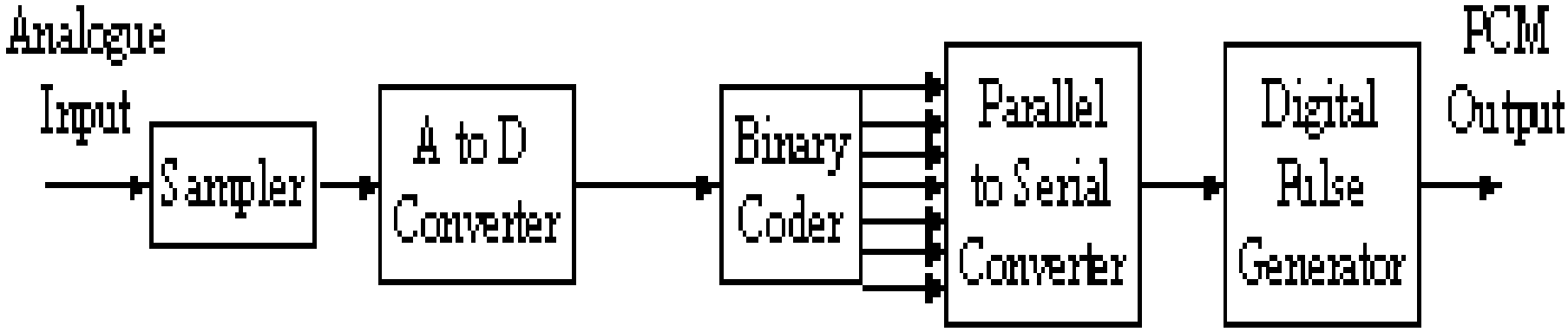


Proses Coding digital

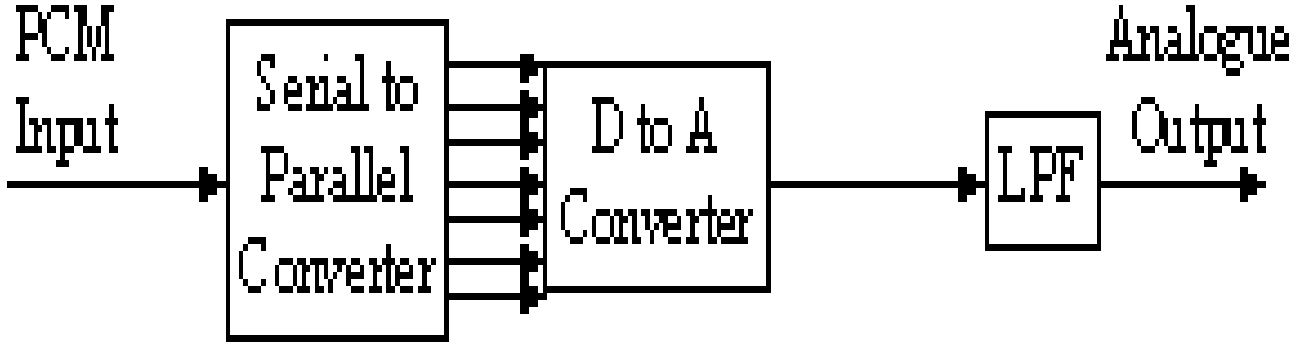


Proses Decoding

Modulator



Demodulator

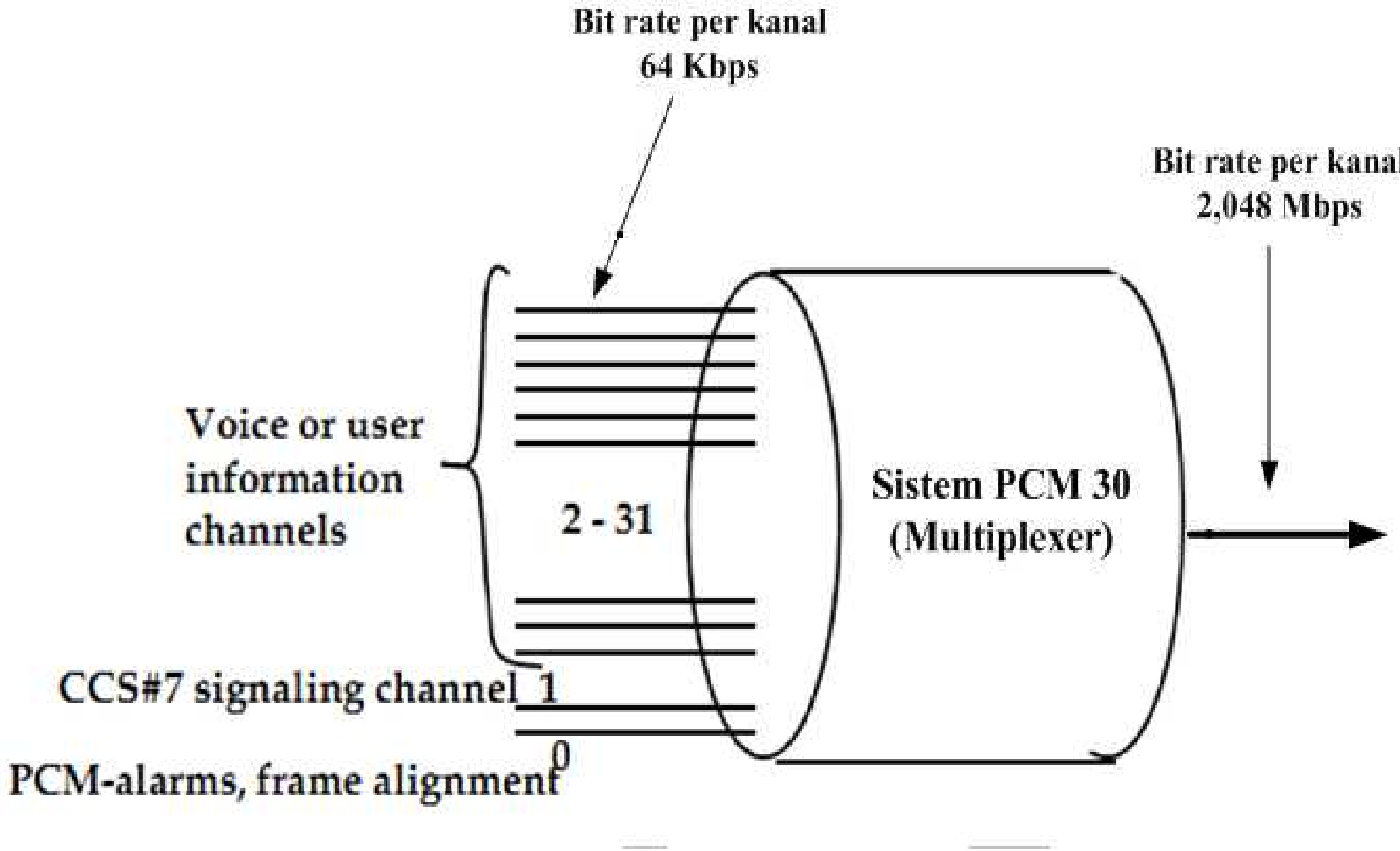


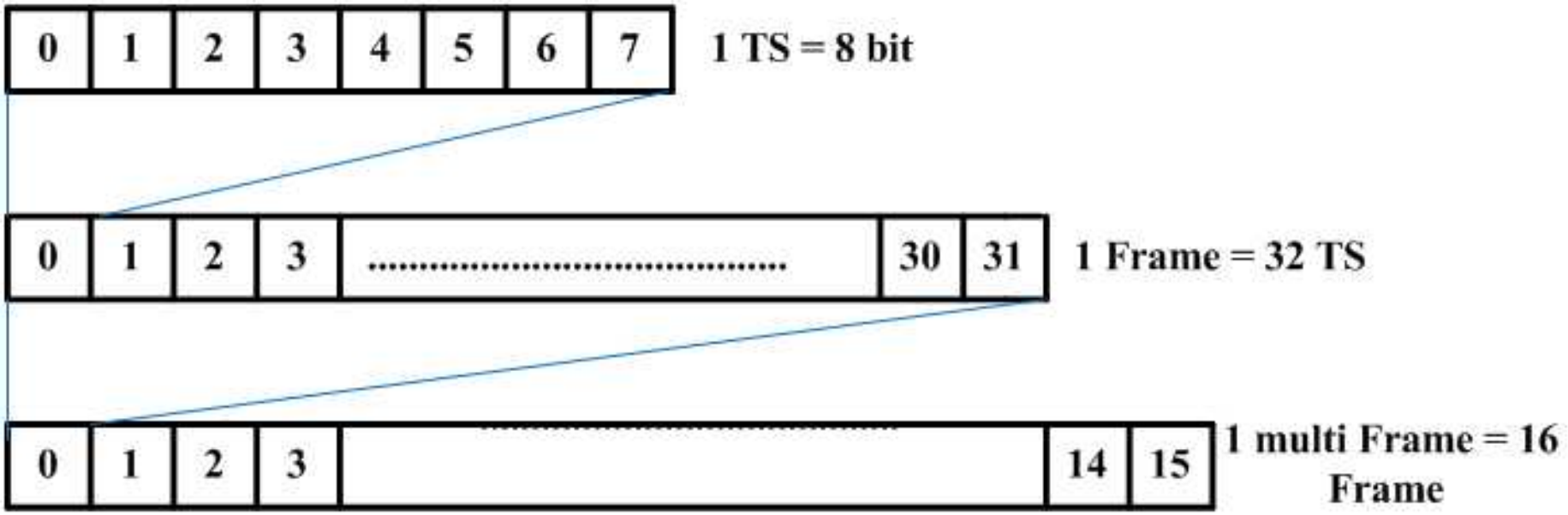
❖ Susunan Frame PCM

Sistem PCM : 1. PCM 30 : Eropa, Indonesia
2. PCM 24 : Amerika Utara (US, Canada), Jepang

▪ Susunan Frame PCM 30 :

- **Jumlah kanal yang dimultiplek 30 kanal suara**
- **Terdapat 32 Time Slot (TS) per frame (30 voice + 2 kanal untuk framing dan signalling**
- **Panjang word (1 TS) sebesar 8 bit (1 sinyal PAM terkuantisasi dikodekan menjadi 8 bit)**
- **Frekuensi sampling fs = 8000 Hz**
- **Bit rate (laju bit per detik) total = 2,048 Mbps**
- **Kompanding yang digunakan : A-Law**


Bit rate per kanal
64 Kbps
Bit rate per kanal
2,048 Mbps
Voice or user
information
channels
2 - 31
Sistem PCM 30
(Multiplexer)
CCS#7 signaling channel 1
PCM-alarms, frame alignment 0

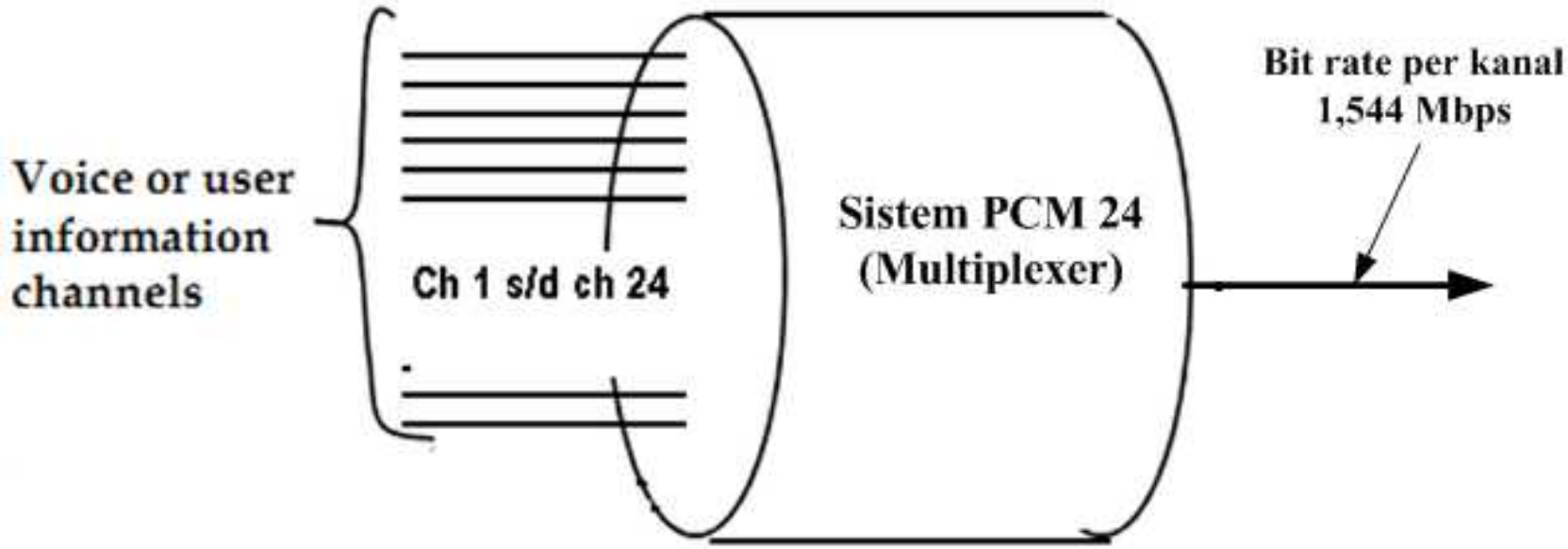


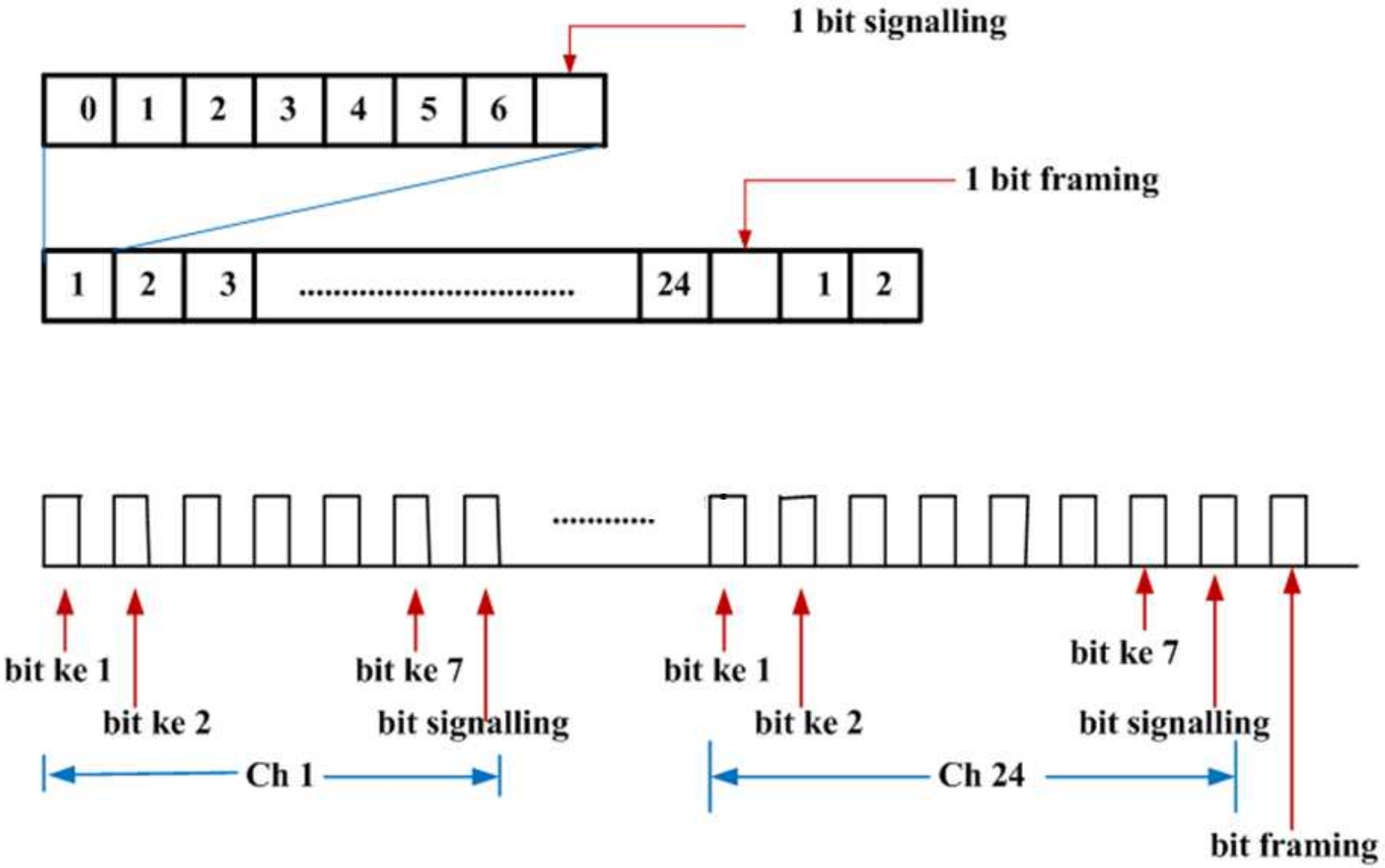
TS no 0 untuk framing
TS no 1 s/d no TS 15 untuk voice
TS no 17 s/d TS no 31 untuk voice
TS no 16 untuk signalling

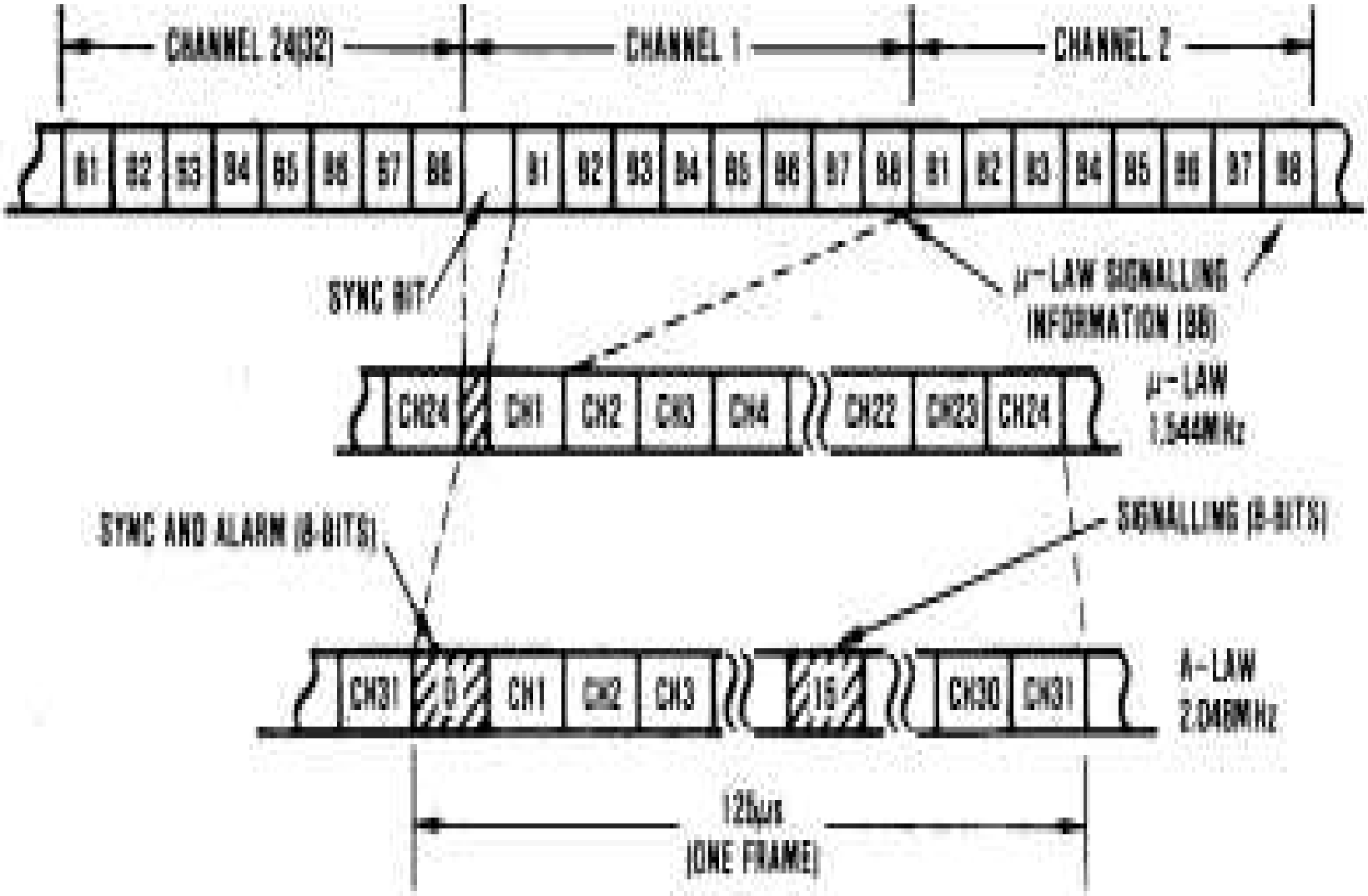
Susunan Frame PCM 30

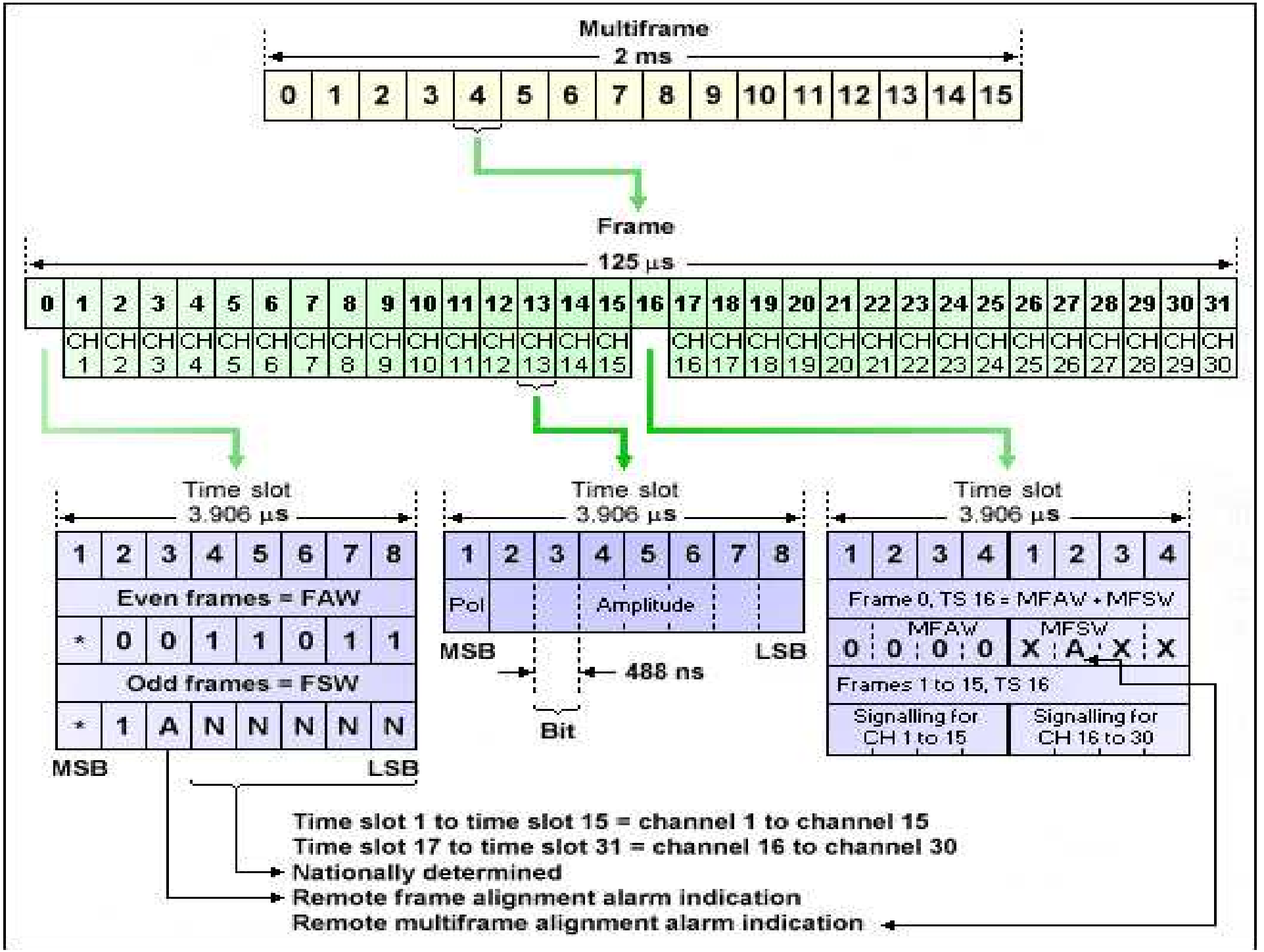
▪ Susunan Frame PCM 24 :

- **Jumlah kanal yang dimultiplek 24 kanal suara**
- **Terdapat 24 Time Slot (TS) per frame**
- **Frekuensi sampling fs = 8000 Hz**
- **Bit rate (laju bit per detik) total = 1,544 Mbps**
- **Kompanding yang digunakan : μ -Law**
- **Jumlah bit per frame 193 bit {(7 bit informasi + 1 bit signalling) x 24} + 1 bit framing**



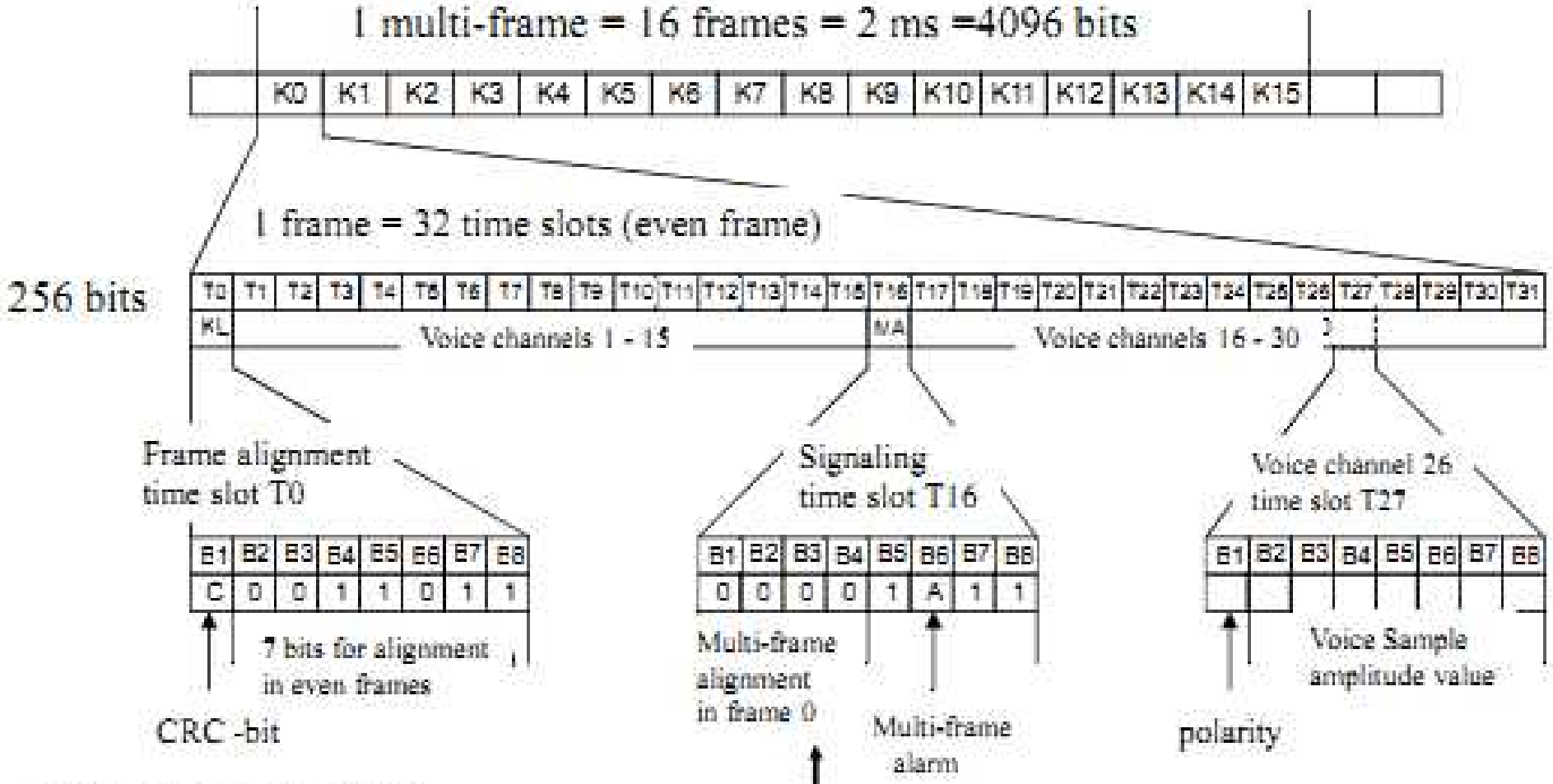

1 bit signalling
0
1
2
3
4
5
6
1 bit framing
1
2
3
.....
24
1
2
bit ke 1
bit ke 2
bit ke 7
bit signalling
Ch 1
bit ke 1
bit ke 2
bit ke 7
bit signalling
Ch 24
bit framing


CHANNEL 24(32)
CHANNEL 1
CHANNEL 2
B1 B2 B3 B4 B5 B6 B7 B8
SYNC BIT
μ-LAW SIGNALLING
INFORMATION (B8)
CH24 CH1 CH2 CH3 CH4 CH22 CH23 CH24
μ-LAW
1.544MHz
SYNC AND ALARM (8-BITS)
SIGNALLING (8-BITS)
CH31 0 CH1 CH2 CH3 15 CH30 CH31
A-LAW
2.048MHz
125μs
(ONE FRAME)


Multiframe
2 ms
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Frame
125 µs
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31
CH 1 CH 2 CH 3 CH 4 CH 5 CH 6 CH 7 CH 8 CH 9 CH 10 CH 11 CH 12 CH 13 CH 14 CH 15
CH 16 CH 17 CH 18 CH 19 CH 20 CH 21 CH 22 CH 23 CH 24 CH 25 CH 26 CH 27 CH 28 CH 29 CH 30
Time slot
3.906 µs
1 2 3 4 5 6 7 8
Even frames = FAW
* 0 0 1 1 0 1 1
Odd frames = FSW
* 1 A N N N N N
MSB
LSB
Time slot
3.906 µs
1 2 3 4 5 6 7 8
Pol
Amplitude
MSB
LSB
488 ns
Bit
Time slot
3.906 µs
1 2 3 4 1 2 3 4
Frame 0, TS 16 = MFAW + MFSW
MFAW
0 0 0 0
MFSW
X A X X
Frames 1 to 15, TS 16
Signalling for CH 1 to 15
Signalling for CH 16 to 30
Time slot 1 to time slot 15 = channel 1 to channel 15
Time slot 17 to time slot 31 = channel 16 to channel 30
Nationally determined
Remote frame alignment alarm indication
Remote multiframe alignment alarm indication

Even numbered PCM 30 -frame

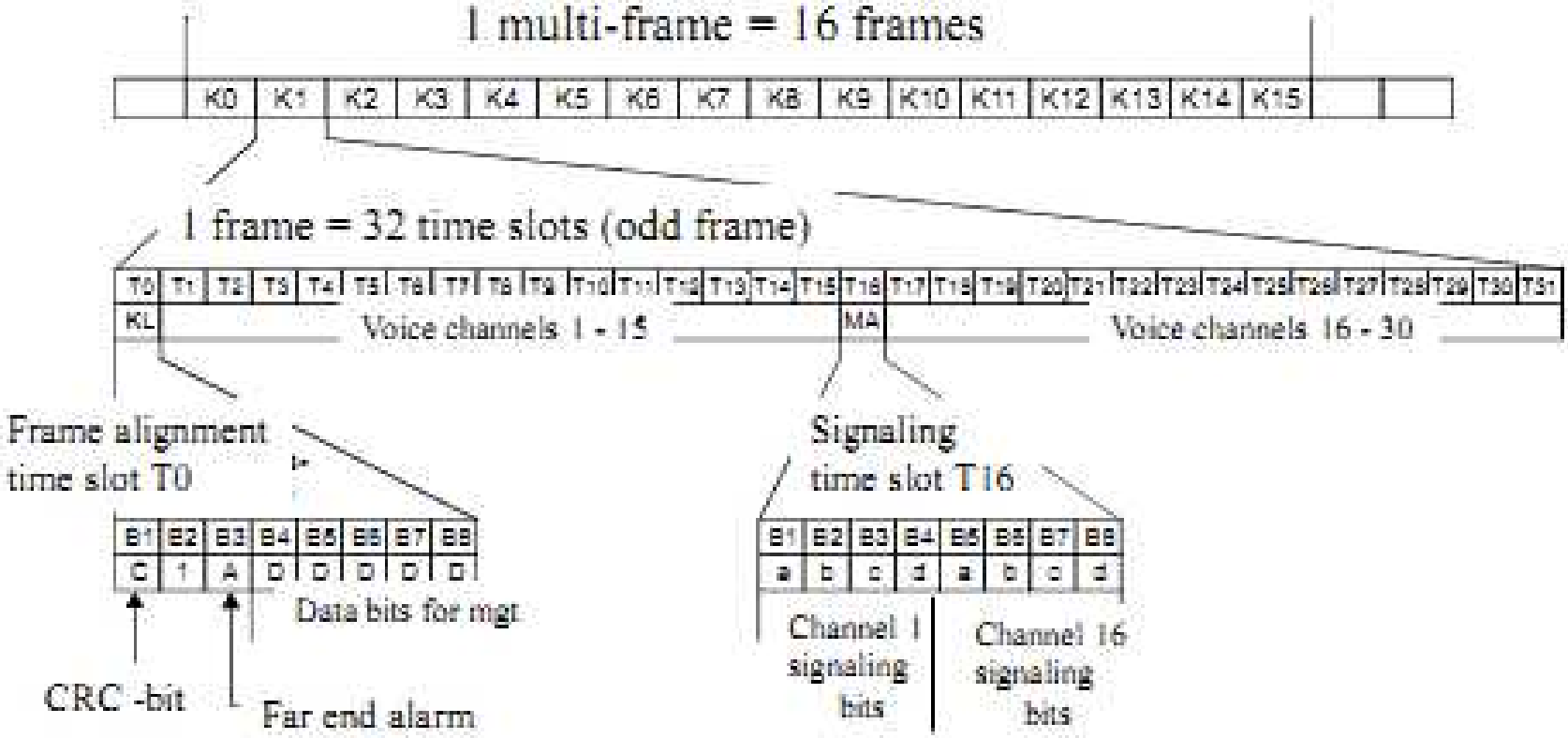
Speed is 2,048 Mbit/s



C bit carries a checksum calculated over previous several frames

Applies only to K0, other even numbered, look at the next slide

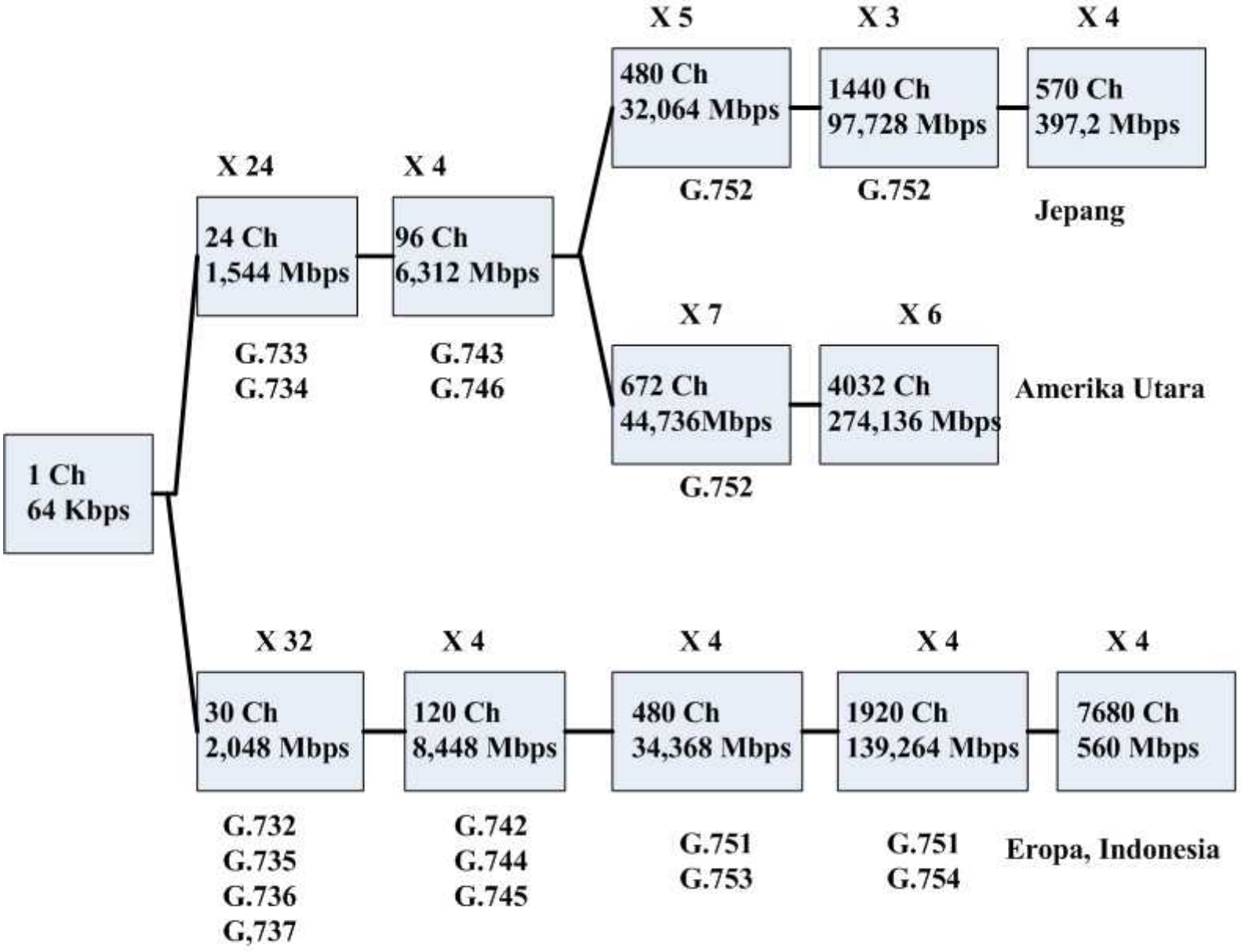
PCM-frame structure (odd frame)



D- bits can be used for managing the PCM system

Actually, this system has two independent multi-frame structures. One based on TSL 16, the other is based on the CRC bit: the first half of the multiframe (2048 bits) is used to calculate the 8-bit CRC and the second half of the multiframe is used to carry the CRC. Nowadays, most times the TSL 16 multiframe structure is historical only.

❖ Hierarhi Sistem PCM :



Hierarhi Pemultiplek PCM 30 dan PCM 24

Terimakasih saudara-saudara

CRC-4 multiframe structure

	Sub-multiframe (SMF)	Frame number	Bits 1 to 8 of the frame							
			1	2	3	4	5	6	7	8
Multiframe	I	0	C_1	0	0	1	1	0	1	1
		1	0	1	A	S_{a4}	S_{a5}	S_{a6}	S_{a7}	S_{a8}
		2	C_2	0	0	1	1	0	1	1
		3	0	1	A	S_{a4}	S_{a5}	S_{a6}	S_{a7}	S_{a8}
		4	C_3	0	0	1	1	0	1	1
		5	1	1	A	S_{a4}	S_{a5}	S_{a6}	S_{a7}	S_{a8}
		6	C_4	0	0	1	1	0	1	1
		7	0	1	A	S_{a4}	S_{a5}	S_{a6}	S_{a7}	S_{a8}
	II	8	C_1	0	0	1	1	0	1	1
		9	1	1	A	S_{a4}	S_{a5}	S_{a6}	S_{a7}	S_{a8}
		10	C_2	0	0	1	1	0	1	1
		11	1	1	A	S_{a4}	S_{a5}	S_{a6}	S_{a7}	S_{a8}
		12	C_3	0	0	1	1	0	1	1
		13	E	1	A	S_{a4}	S_{a5}	S_{a6}	S_{a7}	S_{a8}
		14	C_4	0	0	1	1	0	1	1
		15	E	1	A	S_{a4}	S_{a5}	S_{a6}	S_{a7}	S_{a8}

Note 1 – E = CRC-4 error indication bits (voir le § 2.3.3.4).

Note 2 – S_{a4} to S_{a8} = Spare bits (see Note 4 to Table 4a/G.704)

Note 3 – C_1 à C_4 = Cyclic Redundancy Check-4 (CRC-4) bits (see §§ 2.3.3.4 and 2.3.3.5).

Note 4 – A = Remote alarm indication (see Table 4a/G.704).